

What Capacity Buys at Sub-1M: A CCD-Mixer Study of Attention-Free Language Models

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Abstract

Sub-million-parameter language models are a regime for controlled measurement: at this scale every architectural, recipe, and measurement choice can be held fixed across a full scaling panel, so what capacity buys can be isolated from confounds that larger studies must average over. In that regime we ask: how far can a byte-level language model go with fewer than one million parameters and no attention, recurrence, or state-space machinery? We answer with a controlled six-budget study (1M down to 10K parameters) of one frozen attention-free mixer (V87, the CCD-Mixer introduced here) against one deliberately boring vanilla transformer reference (V88), trained and evaluated under byte-identical recipes. Under this resource allocation the transformer wins raw language modeling at every size and raw memorization at $\geq 500K$ (V88 653.3 versus V87 563.7 full-988 exact match (EM) — verbatim retrieval of trained answers on held-out paraphrases — at 1M, three fine-tuning seeds). The mixer wins where the transformer reference cannot. It wins anti-collapse chatter fluency on the discord conversational register (the pretraining distribution; 0.912 distinct-2 at 500K, the family record). It wins mid-size robustness at 300K (EM 56.3 versus 42.0). It wins the out-of-distribution (OOD) boundary under collapse-inclusive rescoring (V87 39.0% versus V88 57.6% at 1M, 3-seed casual battery). Capacity buys the four capability axes in a tendency: fluency first, then memorization, then boundary. The architecture term is the smallest of the three levers this series investigates; Part 2 covers recipe, Part 3 covers data.

The CCD-Mixer binds four dilated views of one shared depthwise kernel through a per-position content gate. A uniform-1/4 control that freezes the gate to a constant linear time-invariant (LTI) mixture (§7.1a) shows the multi-dilation receptive field, not content-dependent selection, is the primary anti-collapse ingredient: an input-independent uniform mixture at full receptive field matches CCD on cycle prevention and language preservation. The content gate is a measured refinement that lifts chatter fluency by $\approx 0.05 d_2$ and small-budget memorization by localization.

The +19.4 EM lift above the mixer’s immediate parent chassis (V76-RPG) is suggestive rather than significant (three fine-tuning seeds on one seed-42 backbone; Welch $p \approx 0.08$; see §6.1). Limitation: the discord-1M cell is now reported at three pretrain seeds (§6.8), but every other cell remains backbone $n = 1$ (one pretrain seed; only the fine-tuning adapter is 3-seed). External calibration: a WikiText-103 byte-PPL anchor on the pretrained backbones is reported post-registration (§8.13) — on that formal register the raw-LM ordering inverts in the mixer’s favor — and Part 3 carries an official TinyStories validation-split perplexity anchor [Lee, Part 3 of this series, companion paper].

1 Introduction

The dominant paradigm in language modeling is the transformer. Inside that paradigm, attention is treated as non-negotiable: every byte-token sees every other byte-token via a softmax over query-key

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products, the cost is quadratic in sequence length, and alternative token-mixers are evaluated as approximations to attention’s retrieval behavior. This paper asks a different question, one that does not start from attention as a premise.

The question is: how far can a language model go with fewer than one million parameters if it is built entirely from MLP-Mixer operations, with no attention anywhere? The scale is the method, not the apology: at sub-1M every architectural, recipe, and measurement choice can be held fixed across a full scaling panel, so the question admits a controlled measurement rather than another leaderboard entry.

The constraints are deliberate. Sub-1M parameters means a single byte-level autoregressive language model that fits on a consumer GPU (8 GB VRAM, RTX 4060 Ti class), trains in tens of minutes per size, and serves at the same scale. Byte-level means vocab 256, no tokenizer artifacts, no subword out-of-vocabulary behavior. Attention-free means: no softmax over query-key products, no recurrence, no structured state space. Just depthwise convolutions, pointwise projections, gating, and rotational position embeddings. The fourth-generation MicroMixer project (V9 through V88) is the source of the architecture and the eval battery that this paper reports.

Series framing. This paper is Part 1 of a three-paper series on what capacity, recipe, and data buy at sub-1M. Part 2 covers the recipe term (FMSP, the fine-tuning method used here as a black box) [Lee, Part 2 of this series, companion paper]. Part 3 covers the data term: on the same twelve models, chat-format curation is the first-order variable for capability at the 1M budget, and the architecture term remains bounded (the chatter moat is register-dependent) [Lee, Part 3 of this series, companion paper]. The up-front finding: on this chassis, the architecture term is the smallest of the three levers. Under this resource allocation (§8.8), the transformer wins raw language modeling at every size and raw memorization at $\geq 500K$. The CCD-Mixer wins where the transformer cannot, and the wins are bounded (anti-collapse chatter fluency, mid-size robustness, and the OOD boundary under collapse-inclusive rescoring; full numbers and qualifiers in the Abstract). The rest of this paper is the mechanism, the evidence, and the limits.

Four capability axes organize the evaluation.

Fluency is the ability to produce non-pathological continuations under greedy decoding, measured by distinct-2 on chatter (the project’s 9-prompt open-conversation battery) and by per-epoch validation perplexity.

Memorization is the ability to write the 9,012 training QA pairs into the weights at fine-tune time and retrieve them at generation time, measured by full-988 exact-match (the number of held-out paraphrased questions whose greedy answer matches the reference exactly), averaged over three random seeds. Crucially, the 988 held-out questions are unseen paraphrases of trained questions that share the same trained answers, so this axis measures verbatim answer retrieval on held-out paraphrases, not semantic knowledge or factual generalization (see the precise definition in the §6 intro).

Generalization here means paraphrase-form generalization only: the ability to retrieve a trained answer under a paraphrase of the trained question the model has not seen, measured by teacher-forced paraphrase accuracy, and implicitly scored by the memorization battery’s seed variance. This is form generalization (same trained content, unseen surface phrasing), not semantic generalization to new facts the model was never taught.

Boundary discipline is the ability to refuse to fabricate an answer when the question is unanswerable or out-of-distribution, measured by a question-relevance battery on 100 unseen questions, an OOD battery on 59 prompts, an unanswerable battery on 18 prompts plus 6 controls, and a language-preservation battery measuring post-FMSP perplexity drift on the pretraining corpus.

These four axes are not independent. The headline argument of this paper is that under one fixed fine-tuning recipe (the P05 recipe described in §5), they decouple along the capacity axis in a

tendency (not a strict order; two chatter non-monotonicities are noted in §6.4): fluency at every size down to about 300K (a relative advantage; no discord-register size strictly clears the registered WIN bar of $d_2 \geq 0.90$ with zero cycles — §6.2, §8.3), memorization from about 300K upward, and boundary discipline at the 1M budget only for the mixer and at about 500K and above for the transformer — with the 1M boundary reading itself metric-dependent (the legacy collapse-blind OOD metric does not replicate the mixer’s s42 reading at 3 seeds, Table 5 †; the qrel and collapse-inclusive metrics hold it).

The paper makes five contributions, ordered to lead with the portable lessons.

1. **A mechanistic account of repetition collapse in attention-free causal LMs** (§3.2, §3.3, §3.4). Under the constraints {causal, stationary, deterministic, NoPE}, the next-byte map is an autonomous map on a finite state space of 256^W states, so by the pigeonhole principle every trajectory is eventually periodic: collapse is a structural default here, not a tuning bug (the finite-state framing of §3.2 — a property universal across deterministic greedy finite-context LMs, transformers included, that on this chassis is realized as short-period attractors). Rotation-based stationarity breaking (RoPE) works because it makes the map non-autonomous and so removes the periodicity guarantee. On the coherent endpoints, removing all rotation moves EM from 544.3 to 547.3 (+3.0, numerically tied across 3 FMSP seeds; SD not registered), so rotation is not an “unavoidable memorization tax” at the chassis level; its EM effect is background-dependent and non-additive (the full 512 (Nc) versus 564 (NcNg) content-clean conditional contrast is detailed in §7.3). The content-dependent gate does not break orbits on its own: the gate is still an autonomous map, so it is still eventually periodic, and explicit time-dependence (RoPE) is the only non-autonomous forcing in the block. The uniform-1/4 control of §7.1a sharpens the attribution further: the multi-dilation receptive field is the primary anti-collapse ingredient (an LTI uniform mixture at full RF matches CCD on cycle prevention and language preservation), and the content gate is a measured refinement that lifts chatter d_2 by ≈ 0.05 and small-budget EM (300K: 60 vs 24) by localizing precision-critical positions. The gate is real and measured (Table 13), but it is not the collapse-breaker (the V83-NcNg negative control confirms the gate without rotation regresses chatter to 0.733).
2. **A measured chatter↔memorization conservation pattern, and the lift above it** (§3.4, §3.5). On this chassis the two axes can be traded but not exceeded: the registered endpoints sit on a single frontier. The non-LTI mechanism reported here is the first arm holding all four capability axes that is above that frontier, and the lift is carried by the EM axis: +19.4 EM across 3 FMSP seeds (563.7, sd ≈ 4.5 , per-seed 564/559/568, versus 544.3, sd ≈ 11.2 , per-seed 540/557/536). The lift is suggestive rather than significant at $n = 3$ (two-sided Welch $t = 2.78$, $df \approx 2.6$, $p \approx 0.08$; $\alpha = 0.05$ not cleared), but every V87 seed beats every V76 seed by ≥ 2 EM (min gap $559 - 557 = 2$). The chatter-axis reading (+0.011, distinct-2 0.883 versus 0.872, $n = 1$ backbone, seed 42) is a single-seed reading on a 9-prompt battery with no registered confidence interval, so we do not claim a chatter-axis lift — the frontier lift is established on EM as a suggestive lift, not a significant one, and not on chatter. The frontier is measured on one chassis (V76 and its derivatives) with a handful of registered endpoints, so “conservation pattern” denotes a measured trade-off frontier on this chassis, not a universal constraint.
3. **Capability-ordering patterns consistent across both architectures we tested ($n = 2$), from a controlled six-budget scaling study** (§6). One frozen mixer against one deliberately boring vanilla transformer, byte-identical recipes, three FMSP seeds, and the full capability battery at 1M / 500K / 300K / 100K / 50K / 10K parameters. Capacity buys the axes in a tendency (not a strict order; the chatter axis carries two non-monotonicities, §6.4): fluency

first, then memorization with a compounding cliff between 500K and 300K and an EM=0 floor at $\leq 100K$, then boundary discipline. Above 300K the attention advantage is band-limited to mid-range positions, and the residual EM gap is a mechanism gap, not a capacity gap.

4. **An attention-free token-mixing primitive with a perfect-ablation property** (§4, §7.1). The CCD block gates four dilated views of one shared depthwise kernel at every position, adding 517 parameters per block at $d = 128$ and zero extra convolution weights, producing a non-LTI (time-varying) effective filter whose response is a function of the input. Setting `force_dilation=0` reduces the block bit-identically to its RoPE-pattern parent chassis (asserted in the self-test, `diff == 0.0`), so every claim about the gate’s effect is causally attributable rather than inferred from ablation-by-removal. Trained-from-scratch fd0 controls (§4.6) show the gate’s advantage *grows* as the budget shrinks: at 500K/300K the fd0 (LTI) chassis collapses on chatter (5/9 and 9/9 cycles) and loses post-FMSP language preservation roughly 4–5 $\times$ under identical FMSP, at matched pretrain PPL (3.30/3.41 versus 3.29/3.41).
5. **A post-registration audit trail that corrects its own headline numbers** (§5, §6, §7.1, §8). Registered headline metrics, three-seed FMSP protocols, perfect-ablation self-tests (22/22 on the mixer, 20/20 on the reference), and four post-registration audits that re-measure the paper’s own claims when re-measurement disagrees: the collapse-blind OOD inversion (the 500K boundary advantage was a collapse artifact: the collapse correction exceeds the measured gap by a factor of about 2.1, a 37.8pp swing on an 18.1pp measured gap — the 3-seed mean gap V87 35.0% minus V88 16.9%, per [OOD_COLLAPSE_RESCORE_ANALYSIS.md](#) §1; the s42 single-seed gap is 23.7pp), the register-dependent chatter advantage (the reference clears the WIN bar on SmolTalk2), and the fails-loudly-vs-fails-fluently safety asymmetry (the V87-500K hijacker emits $\sim 52\%$ clean verbatim facts under wrong keys vs the V88-500K $\sim 8\%$, making the mixer’s failure mode epistemically the more dangerous one).

The paper is part 1 of a three-paper series. Part 2 develops the FMSP (Fine-tuning with Minimal Parameter changes for Small-parameter LMs) method that this paper uses as a black box [Lee, Part 2 of this series, companion paper]. Part 3 studies pretraining-corpus choice on the same twelve models and shows that chat-format curation is the first-order variable for capability at the 1M budget (the architecture term remains robust at $\leq 500K$) [Lee, Part 3 of this series, companion paper].

Scope. This is a study at the sub-million-parameter scale, where every architectural, recipe, and measurement choice can be held fixed across a six-budget panel: 996,873 parameters at the largest, 200K conversational turns and 9,012 QA pairs, three pretraining epochs and ten fine-tuning epochs per seed. That scale is the point: controlled measurement at a scale where every confound can in principle be eliminated is exactly what a study of what capacity buys requires, not a state-of-the-art apology.

1.1 Glossary of project terms

This paper reuses a small set of project-internal terms (mirroring Part 2’s glossary) that have precise meanings narrower than their everyday senses:

- **Registered.** A metric or run whose value was committed to the public record (the README / log / this paper) before the post-registration analyses began; registered numbers are not edited by the post-registration audits, only re-contextualized.

- **Chatter.** The 9-prompt open-conversation battery decoded greedily on the pretrained backbone; the reported metric is mean distinct-2 (d_2) and the cycle count. The WIN bar is $d_2 \geq 0.90$ with zero cycles.
- **Echo / hijack / refuse.** The three qrel classes on the 100-question question-relevance battery: echo = the correct memorized answer to an unseen paraphrase; hijack = a verbatim memorized answer emitted to a question whose answer the model does not know; refuse = the model declines to answer. echo + hijack + refuse = 100.
- **OOD hijack.** A verbatim memorized answer emitted to an out-of-distribution (non-QA-corpus) prompt on the 59-prompt OOD battery.
- **Zoning.** (Part 2 term, referenced but falsified here.) The hypothesis that memorization and boundary discipline could be separated by partitioning parameters into zones; falsified at every zone size in the V78 experiment.
- **Collapse-inclusive.** An OOD-hijack rescore rule (probe A: any word 3-gram repeating $\geq 5\times$ = collapsed = reclassified as hijack) that flags periodic-orbit collapses the legacy detector scores as “free.” Contrast with **collapse-blind** (the registered legacy detector).
- **Frontier.** The chatter $\leftrightarrow$ EM Pareto frontier of registered V76-chassis endpoints; “above the frontier” means Pareto-dominating a registered endpoint on both chatter and EM (§3.4), not lying above a linear interpolation.
- **fd0.** The trained control that forces the CCD gate one-hot on the dil=1 branch (force_dilation=0); LTI and receptive-field-shrunk to 128 positions. At matched weights, V87-fd0 forward is bit-identical to V76-RPG.
- **uniform-1/4.** The control of §7.1a that freezes the CCD gate at the constant uniform mixture 1/4 on all four branches; LTI but receptive-field-preserved. Isolates content-selection (removed) from receptive field (preserved).

The nomenclature table below collects the project-internal terms used in this paper in one line each; the glossary of §1.1 gives the longer definitions.

2 Related Work

MLP-Mixer and gMLP. The original MLP-Mixer applied transposed pointwise and depthwise operations to patches of an image [Tolstikhin et al., 2021]. The follow-up gMLP added a spatial gating unit that let the depthwise branch learn position-dependent behavior [Liu et al., 2021]. Both showed that, on vision tasks at fixed capacity, the all-MLP baseline could match transformer-quality results. The CCD-Mixer inherits the depthwise-then-pointwise macrostructure but swaps the depthwise spatial convolution for a depthwise causal dilated convolution on a 1D byte stream, and replaces the gMLP gating unit with a softmax over four dilation choices driven by per-position content.

Hyena and long convolutions. Hyena replaced the token-mix of a transformer block with a stack of long depthwise convolutions interleaved with elementwise gating [Poli et al., 2023]. The CCD-Mixer is in this lineage but compresses the long-convolution structure into one shared kernel at four dilations and adds the content gate that makes the token-mix non-LTI (content-dependent). The genesis section traces how the project’s V71 step was a Hyena-style gated long convolution with kernel length 129.

Nomenclature. The project-internal terms used in this paper, in one line each; the glossary of §1.1 gives the longer definitions.

term	meaning
<i>Version and arm tags</i>	
V-numbers (V9–V88)	design notes archived in the project’s public repository; a V-tag names an experiment or design note, not a publication
RPC / RPG (V75 / V76-RPG)	RoPE on the conv value pathway only (RPC) or on both value and gate pathways (RPG); V76-RPG is the CCD-Mixer’s parent chassis
V83-RPG / V87	the V86-census champion: the CCD block on the V76-RPG chassis; V87 freezes it verbatim (arm parameter removed, ctx 1024) at six budgets, and the V87-1M row reproduces V83-RPG to the last digit
V88	the vanilla transformer reference (RoPE q/k , RMSNorm, GELU MLP, no biases, tied byte head, $d_{\text{head}} = 16$); model-card name MicroT-test1
census arms V46-3 / V52 / V58 / V59	old-era mixer families re-run under P05 in the V86 1M census
NcNg / GCR (V82)	rotation-removal contrast arms (GCR = gate-content-rotation skip); NcNg = minimal rotation, both content halves clean
NoRemix (V85)	the sidecar-deleted ablation; shows the sidecar is OOD insurance
<i>Recipe and method tags</i>	
FMSP	Fine-tuning with Minimal Parameter changes for Small-parameter LMs (Part 2); used in this paper as a black box
P05	the registered FMSP recipe: 5% Fisher-guided per-scalar true freeze, answer-only CE, probe-KL, rank-16 zero-init adapter, WSD (warmup-stable-decay) schedule, 10 epochs
fd0	trained control with <code>force_dilation=0</code> (gate one-hot on dil=1); LTI, receptive field shrunk to 128 positions; bit-identical to V76-RPG at matched weights
uniform-1/4	trained control with the gate frozen at the constant 1/4 mixture on all four branches; LTI but receptive-field-preserved
s42 / s43 / s44	FMSP seed shorthand (seeds 42/43/44); backbone pretraining is seed-42-only

FNet and Fourier mixing. FNet replaced attention with a fixed 2D Fourier transform along the sequence axis and showed that this fixed non-learned mixer preserved most of the encoder’s downstream performance [Lee-Thorp et al., 2022]. The CCD-Mixer shares FNet’s spirit (attention is not necessary) but rejects the FFT choice because Fourier mixing is non-causal by default.

Synthesizer. Synthesizer explored learned token-mixes that do not depend on query-key content [Tay et al., 2021]. The CCD-Mixer is content-dependent (the gate reads the block input), but unlike attention the gate does not route values to keys. The gate selects between four conv outputs rather than between four value trajectories.

Byte-level causal dilated convolutions. ByteNet [Kalchbrenner et al., 2016] introduced byte-level (character-level) causal dilated convolutions for neural machine translation with a dilation schedule that grew the receptive field efficiently, and WaveNet [van den Oord, A. et al., 2016] extended causal dilated convolutions to raw audio waveform generation with a stacks-of-dilations schedule. Both belong to the direct lineage of causal dilated convolution on a byte/sample stream that the CCD-Mixer inherits — the CCD token-mix is a byte-level causal dilated convolution at the same granularity as ByteNet and WaveNet, applied at four dilations of one shared kernel. The crucial difference is that ByteNet and WaveNet use a fixed dilation schedule with separate kernels per dilation (and per layer), whereas the CCD-Mixer applies ONE shared kernel at four dilations and binds them through a per-position content gate; the gate makes the effective filter non-LTI in a way a fixed-stack dilated convolution cannot.

Conditionally parameterized and dynamic convolutions; mixture-of-experts routing.

Nomenclature (continued).

term	meaning
<i>Architecture terms</i>	
CCD / CCD-Mixer	content-gated dilated convolution: one shared $k = 129$ depthwise kernel at dilations (1, 2, 4, 8) bound by a per-position 4-way content gate (+517 params per block)
sidecar / ReMixerLayer	question-anchored $K=4$ -head causal weighted-mean pool frozen at the last “Assistant:” marker, injected through a zero-init rank-16 GELU adapter (5,014 params per block)
non-LTI	the effective filter is input-dependent (time-varying), not linear time-invariant
tied byte head	logits computed as $\text{out_norm}(h) \cdot E^T$ against the byte embedding; no separate <code>lm_head</code>
<i>Metrics and batteries</i>	
(eval) battery	the fixed probe suite: chatter, full-988 EM, qrel, OOD, fabrication, language preservation
full-988 EM	exact-match count on 988 held-out paraphrases of trained questions (verbatim answer retrieval); 3-FMSP-seed mean
qrel (echo / hijack / refuse)	the 100-question question-relevance battery: echo = correct memorized answer to an unseen paraphrase, hijack = verbatim memorized answer to a question whose answer the model does not know, refuse = declines
chatter d_2 / cycles	mean distinct-2 on the 9-prompt open-conversation battery; cycles = how many of the 9 prompts trip the loop detector
WIN bar / EM bar	registered pass thresholds: chatter $d_2 \geq 0.90$ with 0 cycles; full-988 EM ≥ 455
fabrication X/18	fabricated answers among 18 unanswerable questions (plus 6 controls)
Discord PPL / qa-1k TF	post-FMSP perplexity on Discord-Dialogues (against the 3.31 chatter-battery baseline);
PPL	teacher-forced perplexity on the 1,000-question held-out qrel extension
FT% / WO	teacher-forced answer-byte accuracy / word overlap with the reference answer
<i>Status and reading terms</i>	
registered / analysis note	committed to the public record before the post-registration analyses began / a post-registration companion analysis that re-measures registered claims without changing them
arm / champion	a named experimental configuration in the project registry / the V86-census winner (V83-RPG)
register / moat	the style of the training distribution (here casual discord conversation) / the mixer’s chatter advantage on that register, which is register-dependent, not universal
cliff / EM=0 floor / 300K inversion	the scaling phenomena: compounding EM drop between 500K and 300K; EM exactly 0 at $\leq 100K$; at 300K the mixer beats the transformer on EM (56.3 vs 42.0), inverting the $\geq 500K$ ordering

Input-dependent combinations of shared basis kernels have close prior art that the CCD gate directly resembles. CondConv computes a content-dependent weighted sum of expert convolution kernels [Yang et al., 2019]. DynamicConv (Lightweight and Dynamic Convolution) performs input-dependent weighting of a small set of depthwise convolution kernels and is the closest analogue to the CCD gate [Wu et al., 2019]. Dynamic filter networks generate filters conditioned on the input [Jia et al., 2016]. HyperNetworks generate the weights of one network from another [Ha et al., 2017]. Because the CCD gate is a 4-way content-dependent softmax over branches (a soft 4-expert mixture with constrained shared experts), the mixture-of-experts routing line is also relevant prior art: sparsely-gated MoE [Shazeer et al., 2017], Switch Transformer [Fedus et al., 2022], and GShard [Lepikhin et al., 2021] all learn a content-dependent routing over expert sub-networks. The single mechanistic difference between the CCD gate and the dynamic-convolution / MoE-routing family is that the CCD gate binds a content-weighted mixture of dilations of ONE shared kernel at every position, whereas CondConv / DynamicConv / MoE-routing weight separate per-expert or per-dilation kernels; the byte-level autoregressive sub-1M application and the anti-collapse purpose are downstream of that single

mechanism gap, not separate axes of difference. **Untied-arm confound.** An untied 4-kernel variant (four separate per-dilation kernels, content-gated) was not run, so the shared-kernel constraint’s contribution to the measured CCD effect is unmeasured; see §8.14.

State-space models (S4, H3, Mamba). Structured state-space models treat the token-mix as a linear recurrence with structured state matrices [Gu, Goel, and Ré, 2022; Fu et al., 2023]. Mamba made the state-space parameters input-dependent [Gu and Dao, 2023]. The CCD-Mixer is conceptually adjacent: a content-dependent linear operator that mixes tokens. The crucial difference is that the CCD-Mixer makes a finite, bounded (four-way) content-dependent selection rather than projecting the input through a learned state-space matrix, and the operator is non-LTI in the strict sense but does not maintain a hidden state across positions.

Other attention-free / linear-recurrence families. Beyond structured state spaces, several recent lines also remove softmax attention while preserving a long-context mixing path: RWKV casts attention-free mixing as a linear recurrence that can be trained in parallel like a transformer [Peng et al., 2023]; RetNet combines a retention operator with a recurrence decoupling trick [Sun et al., 2023]; Griffin interleaves gated linear recurrences with local attention [De et al., 2024], and RecurrentGemma is its production small-LM instantiation [Botev et al., 2024]; xLSTM extends the classic LSTM with exponential gating and matrix memory [Beck et al., 2024]. Like the SSM family, all of these maintain a hidden state or recurrence across positions, whereas the CCD-Mixer’s token-mix is a stateless, per-position, content-gated selection over dilated views of one shared depthwise kernel.

OAMixer and content-aware mixing. OAMixer introduced an object-aware mixing layer for vision transformers using an exponential similarity mask [Kang, Mo, and Shin, 2022]. That paper is a vision method and does not perform causal byte-level language modeling; the project’s ReMixerLayer is an adaptation of OAMixer’s exponential similarity mask to causal byte-level mixing with byte-identity labels.

Rotary position embeddings (RoPE). RoPE encodes absolute position through rotation of query and key pairs [Su et al., 2024]. The CCD-Mixer uses RoPE on the conv value and gate pathways (the RPG pattern), in the form of non-persistent buffers with 64 channel pairs and the standard $\theta_i = 10000^{-2i/64}$ schedule, adding zero parameters and zero state-dict keys.

SwiGLU. SwiGLU is the gated linear unit variant with a SiLU activation that Shazeer [2020] showed improves transformer language modeling perplexity. The CCD-Mixer’s channel-mix is a SwiGLU with hidden width $h = \lfloor (4d^2 + 2d)/(3d + 2) \rfloor = 170$ at $d = 128$.

Parameter-efficient fine-tuning. The FMSP recipe used in this paper is a parameter-efficient fine-tuning method for sub-1M autoregressive LMs. The methodological ancestry is LoRA [Hu et al., 2022], Housby adapters [Housby et al., 2019], BitFit [Ben-Zaken, Ravfogel, and Goldberg, 2022], FishMask [Zhao et al., 2020], and ChildTuning [Xu et al., 2021]. The cited works evaluate BERT-scale or larger models and establish methodological ancestry rather than direct evidence at this scale.

Knowledge capacity and memorization scaling. The knowledge-capacity work of Allen-Zhu and Li [2024] established scaling laws for how many bits of factual knowledge a transformer can store per parameter. The Carlini et al. [2022] quantification of memorization across language models provides the methodological reference for the full-988 exact-match battery. The empirical finding in §6, that the memorization floor under one recipe is consistent across both architectures we tested ($n = 2$) and sits between 100K and 300K parameters, is consistent with the qualitative shape of those published scaling laws at a regime where they had not previously been measured.

What this paper is not. The CCD-Mixer is not a substitute for Mamba at the long-context regime, it is not a Hyena at scale (the largest model has fewer than 1M parameters), and it is not a vision model. The contribution is the block, the genesis narrative, and the controlled six-budget study; it does not claim leadership on any single benchmark.

3 Genesis: Why the CCD-Mixer Had to Be Born

This section is a compressed narrative arc from the first audit to the V87 freeze. The version numbers cited refer to design notes archived in the project’s public repository; the experimental records behind every claim are the source for the registered numbers in §6.

3.1 The 0.36% problem (experiment V71)

The pre-CCD champion was a content-pool MLP-Mixer (V67) with 996,250 parameters that had reached 149/988 exact-match on the held-out QA battery. A line-by-line code audit (V71) produced a striking statistic: 92.84% of the parameters were pointwise channel operations (924,928 params), and only 3,591 parameters (0.36%) did learned cross-position mixing. The pretrain validation perplexity had been pinned at 3.27 to 3.28 across the V59 through V68 generation. The conclusion was quantitative: backbone capacity for cross-position mixing was the binding constraint inside the 1M budget.

The V71 fix stripped the duplicated pointwise expansion and reinvested roughly 461K parameters in a Hyena-style gated long convolution with kernel length 129 and seven blocks, yielding 994,458 parameters. The result was exact-match 170/988 (a 14% relative improvement over V67’s 149), with pretrain perplexity 3.23 breaking the 3.27 plateau. The missing function was content-dependent global mixing.

3.2 The loop attractor diagnosis (experiments V72 and V75)

The improvement was not free. The V72 backbone, trained with the GELU-to-SwiGLU swap that isolated a +12 exact-match effect at parameter parity, produced fluent-looking text that nonetheless fell into periodic-orbit repetition: “...and you are a silly company and you are a silly company and...” The perplexity was 3.16 and the model had decoded-time repetition defenses, but the attractor survived every knob. The backbone’s distinct-2 score sat at 0.731 to 0.798.

The root cause was structural (V75), and it admits a finite-state formulation that is strictly stronger than the shift-equivariance observation.

Finite-state proposition. A byte-level autoregressive model with vocabulary $V = 256$, finite context window $W = 1024$, and a *deterministic* (greedy), *stationary* (parameter-fixed) next-byte map f defines an autonomous dynamical system on a finite state space of size $V^W = 256^{1024}$. Every autonomous map on a finite set is eventually periodic by the pigeonhole principle (the trajectory must revisit a state, and determinism forces a cycle from the first revisit), so periodic-orbit collapse is a structural guarantee of the autonomous finite-state dynamics on this chassis — not a tuning artifact — *unless* the map is made non-autonomous. The earlier shift-equivariance reading (every LTI mixer operator preserves P-periodic input, so periodic orbits are invariant) is the linear-time-invariant corollary; the finite-state proposition is strictly stronger because it gives *attraction* (eventual periodicity), not merely invariance. (We state “proposition” rather than “theorem” because the cumulative-mean pool is a causal running statistic that is only approximately shift-equivariant on a finite-window periodic input — the conclusion holds empirically here, as the §3.1 collapse reproduces in every LTI-only arm.)

Scope caveat. The proposition above applies to any deterministic stationary finite-context greedy autoregressive language model, transformers included, so the pigeonhole-periodicity guarantee is universal across the family. The proposition therefore cannot itself explain why the attention-free LTI token-mixers in this study are unusually collapse-prone; it only certifies that collapse is a structural possibility for the class. Eventual periodicity with a period of up to $V^W = 256^{1024}$ is

an existence guarantee, not the empirical phenomenon: the observed collapses realize short-period attractors quickly (the period-3 word orbits of §3.1, the 5/9 and 9/9 cycle counts in §4.6), and the proposition does not predict short periods. The empirical content of this section is that LTI token-mixers on this chassis find short attractors quickly; the proposition supplies the finite-state framing for that finding, not an alternative account of it.

The proposition has an immediate consequence for the CCD gate. The content-gated dilation mixture is a deterministic function of the input bytes only, so the CCD gate taken *alone* — without any position injection — is still an autonomous map on the same finite state space, and therefore still eventually periodic. This is the theoretical content of the V83-NcNg negative control (§3.5): with rotation removed, CCD regresses chatter to 0.733, because the gate alone does not make the map non-autonomous. RoPE’s absolute-position rotation $R(t)$ injects the step index t into the next-byte function, so under RoPE the map becomes $f(\text{state}, t)$ rather than $f(\text{state})$ — non-autonomous — and the pigeonhole-periodicity guarantee no longer applies. RoPE is the non-autonomous term that *removes the periodicity guarantee*; it does not make periodicity *impossible* (the non-autonomous term could in principle itself be periodic, and empirically 1–2/9 chatter cycles survive even with full rotation), but it removes the structural guarantee the autonomous version carries. A pool-abort probe raised distinct-2 to 0.860 but introduced new attractors; the pool is a regularizer, not the cause. A uniform phase injection probe was bit-identical to the unmodified forward pass because RMSNorm absorbs any uniform per-position signal; a position signal must be directional and trained to remove the autonomous-map guarantee.

3.3 Zero-parameter RoPE patterns: RPC and RPG (experiments V75 and V76)

The first position signal in the project’s history was rotary position embedding applied to the convolution value pathway v only (V75, the RPC pattern). The implementation used 64 channel pairs with the standard schedule $\theta_i = 10000^{-2i/64}$, non-persistent cos/sin buffers, and zero new parameters. The mechanism: $o_t = R(t) \cdot \sum_k w_k \cdot R(-k) \cdot v_{t-k}$, and the outer $R(t)$ breaks the within-sequence shift-equivariance that sustains $h_t \equiv h_{t-P}$ (so the autonomous-map periodicity guarantee of §3.2 no longer applies; generation periodicity is reduced, not eliminated). A self-test on a period-20 steady state produced $\max |\Delta \logit| = 1.19$ for V75 versus 0.0000 for V72.

The results were a partial success. Distinct-2 [Li et al., 2016] chatter rose from 0.798 to 0.863, the cycle count dropped to 2/9 prompts, and the full-988 exact-match stayed EM-safe (451.3 ± 36 , under the pre-P05 FMSP recipe). The verdict: one rotation site is insufficient. The gate pathway g still carried an unmodulated stationary path into the same $v \odot g$ product.

The fix was to rotate both v and g (V76, the RPG pattern), still with zero new parameters. Distinct-2 reached 0.872, cycles dropped to 1/9, perplexity reached 3.18, and full-988 exact-match reached 457.0 ± 32 (pre-P05 FMSP recipe), passing the project’s ≥ 455 bar. V76-RPG became the recommended backbone.

3.4 The RoPE memorization tax and the conservation pattern (experiment V82)

The P05 fine-tuning recipe (which is taken as fixed throughout this paper) raised the V76 backbone’s full-988 exact-match to 544.3 (per-seed 540 / 557 / 536, $\text{sd} \approx 11.2$ — source: paper2 §4.4 and [V77_README.md](#)). A follow-up experiment (V82) asked whether removing RoPE from one of the two pathways would help: the GCR (gate-content rotation) skip produced a co-champion with exact-match 547.3, numerically tied with the full-rotation chassis ($\Delta +3.0$ across 3 FMSP seeds; SD and a paired test were not registered, so we state “numerically tied,” not “statistically tied”).

Registered on the V76 chassis, the coherent endpoints differ by +3.0 (544.3 $\rightarrow$ 547.3) across 3 seeds, so at the chassis level removing rotation buys a numerically-tied shift, not a large EM gain. The +52 and +104 figures are conditional, same-seed contrasts on a fixed background, not chassis-level taxes (the full 512 (Nc) versus 564 (NcNg) content-clean conditional contrast is analyzed in §7.3; see §7.3 for the additive/non-additive reading). The deeper finding was a conservation pattern (a measured trade-off frontier on this chassis, not a universal law): “V76 (full rotation): chatter 0.872, EM 544.3. NcNg (minimal rotation): chatter 0.791, EM 547.3. You can shift the balance; this chassis does not (yet) let you exceed the sum.” The frontier here is a Pareto frontier of registered V76-chassis endpoints (arms that are non-dominated on the chatter $\leftrightarrow$ EM plane), not a connecting line between the two endpoints; “above the frontier” in this paper means Pareto-dominating a registered V76-chassis endpoint on both chatter and EM, not lying above a linear interpolation.

The mechanistic reading was that RoPE rotation is the anti-collapse mechanism, with a background-dependent EM effect: on the coherent V76 endpoints the cost is only +3.0 (numerically tied across 3 seeds), while on a fixed content-clean background the rotation scatters byte-identity and costs EM in a conditional contrast. The V82 design note closed with the question handed to the next experiment: can a mechanism that provides timescale selection without scattering byte-identity break the sum?

3.5 The CCD-Mixer is born (experiment V83)

The V83 answer was to make the token-mix filter time-varying. A content-gated mixture of dilated convolutions is not linear time-invariant: the effective filter changes with the input. The V83 block reused the V76 shared $k = 129$ depthwise kernel at four dilations (1, 2, 4, 8) and added a per-position four-way gate that takes the block input and softmaxes over the four dilations at temperature τ . The gate adds $4d + 5$ parameters per block: 517 at $d = 128$.

The result was that V83-RPG moved above the conservation pattern, and the lift is carried by the EM axis. The first arm in the project’s history holding all four capability axes that is above the chatter $\leftrightarrow$ EM frontier: full-988 exact-match 563.7 (per-seed 564 / 559 / 568, sd ≈ 4.5 , $n = 1$ backbone, seed 42) versus V76-RPG’s 544.3 (per-seed 540 / 557 / 536, sd ≈ 11.2) — a +19.4 EM mean gain across 3 FMSP seeds (difference of the displayed one-decimal means; the exact seed-level difference is 19.33; the full statistical treatment — Welch vs paired, the variance reading, the per-seed min gap — is in §6.1). The +19.4 EM lift is suggestive rather than significant at $p \approx 0.08$, $n = 3$; the deterministic complement is that every V87 seed beats every V76 seed on EM by at least 2 (min gap = $559 - 557 = 2$ EM); the lift is in the same direction on all three seed pairings, just not large enough to clear $p < 0.05$ at $n = 3$. The chatter-axis reading is distinct-2 0.883 (versus V76’s 0.872), a +0.011 gain on a single seed-42 backbone ($n = 1$) on a 9-prompt battery with no registered confidence interval, so we do not claim a chatter-axis lift; the honest summary is that the frontier lift is established on EM as a suggestive lift (+19.4 mean; min seed gap +2; $p \approx 0.08$ at $n = 3$), not a significant one, and not on chatter. Perplexity was 3.18 unchanged. One seed-scope note: the V87-versus-V88 discord-1M gap, the registered headline architecture comparison of §6, is the comparison now backed by three backbone seeds (§6.8); the V87-versus-V76 lift above remains FMSP-level 3-seed on the single seed-42 backbone.

The negative control is essential. Removing the rotation from the gate pathway on the V83 chassis (the V83-NcNg arm) *regressed* distinct-2 chatter to 0.733, below the V82-NcNg baseline of 0.791. The CCD mechanism is complementary to RoPE, not a substitute for it: the gate alone is still an autonomous map (§3.2) and so cannot escape the collapse on its own. The original design intent was that rotation delivers *position* and CCD delivers *timescale* information through gate selection; the §7.1a uniform-1/4 control later sharpened this attribution — the multi-dilation receptive field is

the primary anti-collapse ingredient, and the content gate is the measured refinement (chatter $+0.05 d_2$, small-budget EM by localization). The two compose, and at the rotated endpoint they jointly exceed what either can achieve alone.

3.6 The census and the freeze (experiments V86 and V87)

The V86 census ran every mixer family under one P05 recipe and three FMSP seeds. V83-RPG was the only arm that held all four axes simultaneously: chatter 0.883, full-988 exact-match 563.7, qrel hijack 4.0%, OOD hijack 6.8%. The challengers each failed exactly one advantage. The old “routing ceiling” was a recipe artifact: every old-era byte-AR family gained hundreds of exact-match under P05.

V87 froze the V83-RPG architecture verbatim as the project’s final chassis, removed the ablation `arm` parameter, and extended the context window to 1024 for all six parameter budgets. The 1M row reproduces V83-RPG to the last digit on every registered axis. This paper reports the V87 results.

3.7 The transformer reference (experiment V88)

V88 registered the attention-based comparison baseline so that any future mixer arm can be scored against vanilla transformer at matched budget without rerunning the transformer. The architecture is deliberately boring: causal multi-head attention with rotary position embedding on q and k only, RMSNorm pre-norm, GELU MLP, no biases anywhere, tied byte head, and $d_{\text{head}} = 16$ across all six budgets. V88 exists to put a number on the gap between vanilla attention and the V87 mixer.

4 Architecture

This section specifies the CCD-Mixer block exactly: the operations in execution order, the parameter formulas, the six presets, the self-test suite, and the perfect-ablation property.

4.1 Block specification

The CCD-Mixer block is the V83-RPG chassis, identical to the V76-RPG chassis except for the content-gated dilation mixture. The block operates on input $x \in \mathbb{R}^{T \times d}$ where T is the sequence length and d is the model dimension. The forward pass is:

```

u      = Linear(d -> 2d)(x)
v, g   = u.chunk(2, dim=-1)                # each d-dim
v      = rope(v)                            # full-RoPE on v
g      = rope(g)                            # full-RoPE on g
for d in (1, 2, 4, 8):
    y_d = F.conv1d(pad_(k-1)*d(v), W_conv, b_conv, dilation=d) # SAME weights
w(t)   = softmax(Linear_dil(x)_t / tau)      # per-position 4-way gate; dil_gate weight
    AND bias zero-init -> uniform 1/4 mixture at step 0
v      = Sum_d w_d(t) * y_d(t)              # time-varying effective filter
out    = W_o(v * g)                        # W_o zero-init (silent at init)

```

The four dilated branches reuse the same depthwise weight tensor W_{conv} and bias b_{conv} at dilations (1, 2, 4, 8), so the kernel span ranges from 128 to 1024 positions at $k = 129$. The per-position gate reads the block input x (post-norm) and softmaxes a learned projection $\text{Linear}_{\text{dil}}(x) \in \mathbb{R}^{T \times 4}$ at temperature $\tau = \exp(\log \tau)$. Both the gate weight and the gate bias are zero-initialized, so the gate reads as a uniform 1/4 mixture at step 0; the output projection W_o is zero-initialized so the token-mix is silent at step 0.

The block is wrapped in a three-residual stack with RMSNorm [Zhang and Sennrich, 2019] pre-norm:

```
x = x + token_mix(RMSNorm(x))          # norm1 before the token-mix; GLCTokenMixCCD
x = x + channel_mix(RMSNorm(x))        # norm2 before the channel-mix; SwiGLU, h =
    swiglu_hidden(d)
x = x + remixer(RMSNorm(x))            # ReMixerLayer sidecar: K=4-head causal
    weighted-mean pool of the normed hidden state, frozen at the last 'Assistant: '
    marker (question-anchored), injected through a rank-16 GELU adapter whose up-
    projection is zero-initialized
```

with a tied byte head: $\text{logits} = \text{out_norm}(h) \cdot E^T$ where E is the input embedding matrix (no standalone lm_head). The vocabulary is 256 (byte-level).

4.2 Shared kernel, zero extra conv parameters

The crucial parameter-counting claim is that the four dilated branches add zero new convolution parameters. A single CausalDepthwiseConv1d(d, k) module with weights $(d, 1, k)$ and a d -dim bias is instantiated once and applied at all four dilations. At $k = 129$ and $d = 128$, the conv module is $128 \times 129 + 128 = 16,640$ parameters per block. The four-way gate adds $d \cdot 4 + 4 + 1 = 4d + 5$ parameters per block: 517 at $d = 128$.

The mechanism this enables is non-LTI token-mixing. A linear time-invariant operator has fixed coefficients; a periodic-orbit attractor is generic on this chassis (the orbit argument of §3.2, supported empirically by every LTI-only arm). By letting the per-position content choose the mixture at every time step, the operator’s effective filter becomes a function of the input. The caveat is that “non-LTI” does not by itself break periodic orbits: on a periodic orbit the gate’s input is periodic, so the per-position gate weights are periodic with the same period, and an autonomous content-gated map remains P-periodic-consistent. Explicit time-dependence (the RoPE rotation, applied to both v and g) is the only non-autonomous forcing in the block. The §7.1a uniform-1/4 control sharpens the attribution: the multi-dilation receptive field (preserved by uniform-1/4 even though its gate is LTI) is the primary anti-collapse ingredient, and the content gate’s measured contribution is a refinement that lifts chatter d_2 by ≈ 0.05 and small-budget EM by localizing precision-critical positions — composed with rotation, this lifts the joint endpoint above the LTI chatter $\leftrightarrow$ EM frontier (the V83-NcNg negative control in §3.5 confirms that the gate without rotation makes chatter worse, not better).

4.3 RoPE pattern, output projection, SwiGLU, and sidecar

The full-RoPE-on-both-rotation-sites pattern is the V76 chassis. The implementation uses 64 channel pairs, with pair i at position t rotated by $t \cdot \theta_i$ where $\theta_i = 10000^{-2i/64}$. The cos and sin tables are non-persistent buffers, so they re-derive per forward pass and never enter the state-dict. The block’s state-dict layout is identical to V76’s, with exactly 21 additional CCD keys in total (7 blocks times 3 keys: dil_gate.weight , dil_gate.bias , log_tau).

The output projection W_o is zero-initialized. The channel-mix is a SwiGLU with hidden width $h_{\text{swiglu}}(d) = \lfloor (4d^2 + 2d)/(3d + 2) \rfloor$, which evaluates to 170 at $d = 128$.

The ReMixerLayer sidecar is an adaptation of the exponential similarity mask from OAMixer (a vision method), applied to causal byte-level mixing with byte-identity labels. The sidecar has 5,014 parameters per block at $d = 128$. Over seven blocks the sidecar adds 35,098 parameters, roughly 3.5% of the 1M model. The pool mechanism is a four-head causal weighted-mean pool of h , frozen at the last “Assistant: ” marker in the input, and injected via a rank-16 GELU adapter with the up-projection zero-initialized.

Table 1: V87 presets. All presets share dilations (1, 2, 4, 8), label dimension 16, pool heads 4, and maximum sequence length 1024. (Source: V87 design note §2 and `model_v87_final.py`.)

preset	d_{model}	blocks	kernel k	params	budget use
v87_final_1m	128	7	129	996,873	99.7%
v87_final_500k	96	6	97	491,742	98.3%
v87_final_300k	80	5	81	292,525	97.5%
v87_final_100k	48	4	65	95,084	95.1%
v87_final_50k	32	4	65	48,684	97.4%
v87_final_10k	16	2	33	9,666	96.7%

4.4 Exact parameter formula

```

swiglu_hidden(d)    = (4*d*d + 2*d) // (3*d + 2)          # = 170 at d=128
token_mix/block     = 3*d*d + d*k + 8*d + 5              # proj_in(2d^2+2d) + conv(dk+d)
                    + W_o(d^2+d) + gate(4d+4) + log_tau(1)
channel_mix/block   = 3*d*h + 2*h + d                    # SwiGLU
remixer/block       = 3*d + 2 + (K+r)*(d+1) + r*d         # norm + kappa + saliency +
                    pool_proj + pool_down + pool_up
per_block           = token_mix + 2*d (norms) + channel_mix + remixer
total               = 257*d + num_blocks * per_block      # embed 256*d + out_norm d

```

where $K = 4$ is the pool head count, $r = \max(4, d/8) = 16$ at $d = 128$ is the pool rank, and k is the conv kernel length. The 1M preset’s parameter count has an exact derivation: 993,254 (the V76-RPG chassis) + 3,619 (the seven blocks times 517 CCD gate parameters) = 996,873. The 10K preset is structural: the tied embedding alone ($256 \times 16 = 4,096$) is 42.4% of the 9,666-parameter budget.

4.5 The six V87 presets

The 1M preset is exactly the V83-RPG champion. Below 100K parameters, the conv kernel shrinks, and below 50K the tied embedding floor and minimum block structure begin to dominate.

4.6 Self-tests and the perfect-ablation property

The V87 implementation ships a 22-test self-test suite arranged as three tests per preset (six presets times three test groups) plus four cross-preset tests. The full 22/22 V87 self-test battery and the 20/20 V88 self-test battery are reproducible from the GitHub release by running `src/model_v87_final.py` and `src/model_v88_transformer.py` respectively (see the Code, Data, and Model Availability section). The per-preset tests cover exact parameter count hard-assert, budget assertion, zero-initialization of W_o and the gate parameters, bit-identical initialization versus a token-mix-zeroed forward pass, finite gradients, gradient flow through the conv and the gate after un-zeroing, and causality under perturbation at positions 130, 200, and 255 in a 256-token sequence (past-logit difference equal to 0.0). The cross-preset tests cover the interface contract, module scan (no attention, no recurrence, no state-space), generation smoke under greedy and sampled decoding, and the perfect-ablation test.

The perfect-ablation property is the single most important architectural claim. Setting `force_dilation = 0` forces the gate to be one-hot on the $\text{dil} = 1$ branch, which makes the V87-1M forward pass bit-identical to the V76-RPG parent with matched weights (asserted, maximum difference equal to 0.0). The gate, when free, produces a difference greater than 10^{-3} . Every number

Table 2: The fd0 trained-control (`force_dilation=0`, trained from scratch) versus the registered CCD-trained V87 arms. fd0 arms are discord-only; full-988 EM is 3-seed (42/43/44); all other fd0 metrics are seed-42. (Source: `logs/fd0/.`)

budget	chassis	pretrain val PPL	chatter d_2 (cycles)	full-988 EM (3-seed mean)	post-FMSP Discord PPL
500K	fd0 (LTI)	3.30	0.760 (5/9)	234.3 (229/227/247)	19.96
500K	CCD (registered)	3.29	0.912 (1/9)	279.3 (270/280/288)	4.00
300K	fd0 (LTI)	3.41	0.579 (9/9)	32.0 (38/28/30)	18.23
300K	CCD (registered)	3.41	0.810 (1/9)	56.3 (60/60/49)	4.47

the project has ever reported for V76-RPG can be reproduced as the `force_dilation = 0` ablation of V87, and every improvement V87 shows over V76 is attributable to the content-gated dilation mechanism alone.

The fd0 trained-control: does the gate generalize under training? The property above is asserted at the 1M corner — with *matched weights*, the fd0 forward is bit-identical to the V76-RPG parent. The open question is whether the content gate’s benefit survives when an fd0 (LTI) chassis is *trained from scratch* at smaller budgets, exactly where the §6 scaling study shows the LTI-attractor pressure is strongest. The control arms are V87 pretrained with `force_dilation=0` on the discord register under the identical V76 recipe (§5.1), then fine-tuned under the identical P05 FMSP recipe (§5.2), and scored against the registered CCD-trained V87 arms at the same two budgets. Pretrain quality is matched by construction: at 500K the fd0 val PPL is 3.30 versus the CCD arm’s 3.29, and at 300K both read 3.41.

The four-point reading:

1. *Anti-collapse, and the contribution widens as budget shrinks.* The fd0/LTI chassis trained from scratch collapses on chatter at both budgets (5/9 and 9/9 cycles), and the collapse deepens as parameters shrink (d_2 falls 0.760 at 500K to 0.579 at 300K), while the CCD-gated chassis holds (0.912 and 0.810, each 1/9 cycles). The anti-collapse delta widens with budget pressure (0.152 at 500K, 0.231 at 300K). This generalizes the perfect-ablation argument beyond the 1M bit-identity corner: when the gate is removed by *training* rather than by weight-matching, the LTI periodic-orbit collapse predicted in §3.2 returns, and returns harder at smaller d . **Causal caveat (resolved in §7.1a):** fd0 removes content-dependence *and* shrinks the receptive field together, so this reading point alone does not isolate which change drives the collapse. The uniform-1/4 control of §7.1a (LTI, RF preserved) matches CCD on cycle prevention (1/9 at both budgets), so the primary anti-collapse ingredient is the multi-dilation receptive field, not content-dependent selection; the content gate’s measured role is the chatter-fluency and small-budget-EM refinement reported in §7.1a.
2. *The gate also helps memorization at these budgets.* At 500K the CCD arm beats fd0 by +45.0 EM (279.3 versus 234.3, 3-seed means) and at 300K by +24.3 (56.3 versus 32.0). This contrasts with the 1M corner, where the §3.4 / §7.3 rotation-removal contrasts were EM-tied (544.3 versus 547.3): below 1M the content gate is EM-positive, not EM-neutral.
3. *Post-FMSP language preservation catastrophically fails on the fd0 chassis.* At identical pretrain PPL and under the identical P05 recipe, the fd0 arms read Discord PPL 19.96 (500K) and 18.23 (300K) against the CCD arms’ 4.00 and 4.47 — a $5.0\times$ and $4.1\times$ degradation respectively

Table 3: V88 transformer presets. (Source: V88 design note §1–§2.)

preset	d_{model}	heads	blocks	FFN	params	budget use
v88_transformer_1m	128	8	5	496	996,736	99.7%
v88_transformer_500k	96	6	4	424	498,528	99.7%
v88_transformer_300k	80	5	4	272	297,680	99.2%
v88_transformer_100k	48	3	3	200	97,872	97.9%
v88_transformer_50k	32	2	3	152	49,888	99.8%
v88_transformer_10k	16	1	2	56	9,808	98.1%

— with qa1k TF PPL rising in lockstep (2.58 and 2.54 versus 1.52 and 1.60). We report this as measured. The consistent-with-the-collapse-account reading is that the small-budget LTI chassis sits closer to the periodic attractor (point 1 and the §3.2 mechanism) and fine-tuning pushes it in; the precise mechanism is flagged as future work.

4. *The fd0 arms’ low OOD hijack is degenerate-pass, not boundary discipline.* On the seed-42 OOD battery the fd0 arms read 1.7% hijack at both 500K and 300K, well below the CCD arms’ 35.6% and 25.4%. This is the degenerate-pass pattern already caveat in §6.1 for the 10K row (a partly-collapsed model matches no memorized answer, so the detector scores it “free”): the fd0-500K chassis runs 5/9 chatter cycles at Discord PPL 19.96, and the fd0-300K chassis runs 9/9, so their low hijack is collapse-evasion rather than boundary discipline. The seed-42 qrel battery, which does not reward collapse, reads the fd0 arms *worse* than the CCD arms on echo at both budgets (22.0 versus 37.0 at 500K; 7.0 versus 11.0 at 300K).

Unanswerable fabrication is the one axis the two chassis tie on (fd0 16/18 versus CCD 17/18 at 500K; 16/18 versus 15/18 at 300K), consistent with §6 where fabrication is recipe-determined (no abstention data in P05) rather than chassis-determined.

4.7 The V88 vanilla transformer reference

The V88 architecture is a vanilla pre-norm decoder-only transformer: causal multi-head attention with RoPE on q and k only ($\theta_{\text{base}} = 10000$ with the standard per-pair schedule $\theta_i = \theta_{\text{base}}^{-2i/d_{\text{rot}}}$, non-persistent buffers, SDPA), GELU MLP, no biases anywhere, tied byte head, and $d_{\text{head}} = 16$ at every preset. There is no flash attention, no SwiGLU, no ALiBi, no QKNorm, no sliding window, no multi-query attention, and no mixture of experts. The parameter formula is exact: $\text{total} = 257 \cdot d + N \cdot (4d^2 + 2d \cdot h + 2d)$.

The 20/20 self-test suite verifies exact parameter counts, no bias parameters anywhere, $d_{\text{head}} = 16$ at every preset, RoPE buffers non-persistent across `.to(device)` moves, gradient flow through Q/K/V/O and the MLP, and causality under future-position perturbation.

5 Experimental Setup

5.1 Pretraining recipe (V76)

The pretraining corpus is Discord-Dialogues, 200,000 multi-turn conversational samples in `User:/Assistant:` format (the Hugging Face dataset `mookiezi/Discord-Dialogues`). Sequences are truncated or padded to 1024 bytes. The model is a byte-level autoregressive language model with vocabulary 256 and tied input/output embeddings.

The optimizer is AdamW [Loshchilov and Hutter, 2019] with learning rate 3×10^{-3} , weight decay 0.01, and batch size 16. The learning rate schedule is warmup-stable-decay (WSD) [Hu et al., 2024] with 500 warmup steps followed by a linear decay to zero. The validation split is 10% of the dataset, and the seed is 42. Training runs for 3 epochs with plain cross-entropy on non-pad positions.

5.2 Fine-tuning recipe (P05 / FMSP)

The fine-tuning corpus is 11aa33219/small-qa-en-10k, 10,000 general-knowledge QA pairs split into 9,012 train and 988 held-out paraphrases. The fine-tuning method is FMSP (Fine-tuning with Minimal Parameter changes for Small-parameter LMs), which exists because full fine-tuning destroys sub-1M models. With roughly 2 million bits of learnable capacity, writing 9,012 QA pairs into the weights overwrites the language model itself. FMSP resolves the tension by combining per-tensor freezing of a small fraction of parameters, answer-only cross-entropy, and a probe-KL term that anchors the frozen backbone’s distribution.

The P05 recipe overrides the FMSP defaults as follows: freeze fraction 0.05, true freeze, answer-only cross-entropy, and probe-KL weight 0.5. The adapter rank is 16, the KL term on the QA mixture is 0.05, the learning rate is 1×10^{-3} , the weight decay is 0.01, the Fisher sample size is 200, and the training runs for 10 epochs. Three seeds are used (42, 43, 44) and the reported full-988 exact-match is the mean across the three seeds.

Algorithm (P05, from Part 2). The P05 recipe is a self-contained schematic so that the headline EM numbers in this paper are reviewer-verifiable without Part 2. The full derivation, ablations, and the recipe-search tree are in the companion FMSP paper [Lee, Part 2 of this series, companion paper] and the project’s V77 design note ([V77_README.md](#) §2-§5); the schematic below restates the four moving parts (Fisher-profile freeze mask, zero-init rank-16 adapter, answer-only CE, probe-KL) used to produce every FMSP checkpoint reported in this paper:

```
Inputs: pretrained backbone B (params W_B), QA dataset D, probe corpus P,
        rho = 0.05 (freeze fraction), lam1 = 0.05 (QA KL), lam2 = 0.5 (probe-KL),
        adapter rank r = 16, lr 1e-3, wd 0.01, Fisher sample N = 200, 10 epochs
1. Fisher profile on a 2000-sample subset of the pretraining corpus (Discord-Dialogues),
   not of D: F_j = (1/N) Sum_k ( d log p_B(x_k) / d w_j )^2
2. Freeze mask M = top-rho fraction of scalars by F_j           (49,844 of ~997K at V87-1M
   ; 49,663 of ~993K for the V76-RPG chassis on which P05 was developed)
3. Adapter A = rank-r zero-init up/down on every tensor       (W_o zero-init -> silent at
   step 0)
4. Per step on (D answer-region, probe batch from P):
   L = CE( logits_A(x[a_mask]), a )                             # answer-region only (C1)
     + lam1 * KL( logits_A(qa), logits_B(qa) )                 # QA-mixture KL
     + lam2 * KL( logits_A(probe), logits_B(probe) )           # probe-KL to frozen
   backbone (P05)
   AdamW.step(L); then restore-after-step: w_j <- W_B[j] for j in M # "true
   freeze" (C2)
Output: adapter A on the frozen backbone (4,096 params at V87-1M; <0.5% of backbone)
```

The probe corpus P is 1200 Discord-Dialogues samples plus 800 unseen small-qa-en-1k questions plus 60 fixed weird-OOD prompts (the qrelevance eval questions excluded by normalized-string match), all formatted with the “Assistant: ” cue. Probe-KL is **not** CE-replay: zero CE is computed on probe inputs, so zero memorization capacity is spent on probe content — the term only anchors the trainable 95% of scalars to the frozen backbone’s distribution off-distribution.

5.3 Evaluation battery

The evaluation battery consists of five probes run on the fine-tuned model (full-988 exact-match, question-relevance, OOD, unanswerable fabrication, and language preservation), plus two probes run on the pretrained backbone (chatter and per-position CE).

Chatter (pretrained backbone). Nine prompts are decoded greedily (temperature 0) with repetition penalty 1.2 and no-repeat-4-gram constraint (the repetition-penalty and no-repeat knobs are inert under greedy; see the decode disclosure below), up to 200 tokens. The reported metric is mean distinct-2 and the cycle count. Distinct-2 is a reference-free internal-diversity metric of the generated text (the fraction of distinct bigrams among all generated bigrams), not a direct fluency or quality measure; it correlates with, but does not certify, human-readable fluency. The WIN bar (the registered pass criterion) is mean $d_2 \geq 0.90$ with zero cycles.

Full-988 exact-match (fine-tuned, 3 seeds). The 988 held-out paraphrases are decoded greedily under the same constraints, and the answer is compared to the reference for exact match. The reported metric is the mean across seeds 42, 43, and 44.

Question-relevance battery (fine-tuned, seed 42). 100 unseen questions are decoded under the same constraints, and the output is classified as echo (the model answers with the correct memorized answer to an unseen paraphrase of a trained question), hijack (the model emits a verbatim memorized answer to a question whose answer it does not know), or refuse. The reported metric is the echo/hijack split.

OOD battery (fine-tuned, seed 42). 59 prompts that are out-of-distribution for the QA corpus are decoded under the same constraints and classified as engage or refuse.

Unanswerable fabrication (fine-tuned, seed 42). 18 questions that have no factual answer plus 6 controls are decoded under the same constraints and classified as fabricate, evade, or other.

Language preservation (fine-tuned, seed 42). The post-FMSP perplexity on the Discord-Dialogues validation set is reported as the Discord PPL metric. Note the V76-RPG backbone reads pretrain-validation PPL 3.18 (the val split) but Discord chatter-battery PPL 3.31 (the language-preservation baseline used in Parts 2 and 3); the post-FMSP Discord PPL values reported in this paper (e.g. V87-1M 3.66 in Table 5) are measured against the 3.31 chatter-battery baseline, not against the 3.18 val-split PPL.

Per-position CE probe (pretrained backbones only). A 512-item subset of the validation set (PROBE_A_ITEMS = 512, the first 512 Discord val items) is forward-passed once through the backbone, and the per-position cross-entropy (pre-shifted labels, `labels[t]` = target for position `t`; pad positions excluded via `ignore_index`) is averaged within five position buckets ([0-63], [64-127], [128-255], [256-511], [512-1023]) and overall. Note this probe uses a different position weighting from the pretrain val-PPL number: the probe’s “overall” CE averages over the non-pad positions of a 512-item subset, whereas the reported 3ep val PPL is a token-likelihood-weighted average over the full 10% val split. The two are therefore not expected to reconcile exactly: for V87-1M the probe overall CE is 1.2080 while $\ln(3.18) = 1.1569$ ($\Delta \approx 0.051$), and for V88-1M the probe overall CE is 1.1852 while $\ln(3.08) = 1.1249$ ($\Delta \approx 0.060$) — the gap is in the same direction and of similar magnitude for both architectures, consistent with a subset/weighting difference rather than an architecture effect.

Auxiliary metrics definitions. The “Auxiliary (seed 42)” rows reported under Tables 5 and 6 use three metrics that are computed at the seed-42 FMSP checkpoint and not separately defined elsewhere in this section: **FT%** is the teacher-forced answer-byte accuracy on the train split (the fraction of answer-region positions whose argmax prediction equals the reference byte); **WO** is the word-overlap of the greedy answer with the reference answer (token-set overlap, range 0 to 1); and **qa-1k TF PPL** is the teacher-forced perplexity on the qa-1k probe subset (a 1,000-question held-out

Table 4: Per-axis registered bars.

axis	registered bar / comparison protocol
chatter	WIN bar: mean $d_2 \geq 0.90$ AND zero cycles (no discord-register V87 size strictly passes — §6.2, §8.3; the SmolTalk2 register does pass — §8.3)
full-988 EM	project ≥ 455 bar (inherited from Part 2 / the V86 census best-in-census), 3 FMSP seeds
qrel	echo $\geq 50\%$ AND hijack $\leq 10\%$ (only V87-1M at 95.0/4.0 passes; §6.2)
OOD hijack	collapse-inclusive, 3-seed where available; V87-1M reads 39.0% 3-seed mean (§8.1) — the comparison protocol, not a single threshold, because OOD is a register-pair $\times$ collapse-inclusive property (§8.1–§8.2)

extension of the qrelevance battery, used as a near-distribution language-preservation probe). These three are auxiliary to the four headline axes and are reported only as scaling-trajectory context.

Decode configuration disclosure. Every generation-based metric in this paper (chatter distinct-2, full-988 exact-match, question-relevance, OOD, unanswerable fabrication) is measured under one registered decode configuration: greedy decoding (temperature 0) with repetition penalty 1.2 and a no-repeat-4-gram constraint, applied identically to both architectures (the repetition-penalty and no-repeat knobs are inert under greedy; see the disclosure below). Raw-greedy (penalty-free) numbers are not part of the registered protocol and are not reported here. In this codebase (the registered `generate()` in `src/model_v71_rtm.py`, inherited by V87 and mirrored in V88), both knobs are applied only inside the sampling branch; under temperature 0 the greedy branch takes argmax on unmodified logits, so the registered repetition penalty 1.2 and no-repeat-4-gram are inert. Raw-greedy verification on all 24 pretrained backbones (12 discord + 12 corpus-sweep) produced byte-identical outputs to the registered runs, so the chatter and boundary numbers never had decode assistance; the cross-architecture comparisons remain fair because the configuration is applied identically to the mixer and the transformer. Distinct-2 is a reference-free internal-diversity metric, not a direct fluency or quality measure.

Per-axis registered bars. To make “the first arm holding all four capability axes” (§1, §3.5) a falsifiable rather than a purely relative claim, the registered pass criteria per axis are:

“First arm holding all four axes” is therefore a census-rank claim over these four bars: V83-RPG is the only V86-census arm that is above the chatter $\leftrightarrow$ EM frontier (Pareto-dominating a registered V76-chassis endpoint on both) AND passes the qrel bar AND is not collapse-dominated on OOD. V85-NoRemix is above the chatter $\leftrightarrow$ EM frontier too but fails OOD (§7.2); V52-P05 fails qrel (Table 14).

5.4 Hardware

All experiments were designed for a single consumer GPU (RTX 4060 Ti class, 8 GB+ VRAM). The six-size V87 pretrain ran one model per GPU in parallel on 2026-07-27, with wall-clock times approximately 95 minutes for the 1M model (RTX 4070), 95 minutes for the 500K model, 65 minutes for the 300K model, 45 minutes for the 100K model (RTX 3060 Ti), 35 minutes for the 50K model, and 12 minutes for the 10K model (RTX 3070). The V88 pretrain on 2026-07-28 was approximately 80, 85, 60, 42, 32, and 11 minutes per size. All twelve FMSP checkpoints are available as Hugging Face safetensors repositories under the `11aa33219/` namespace; these twelve released repos are the seed-42 arms (six V87 sizes plus six V88 sizes, discord-register). The 3-seed full-988 EM reported in Tables 5 and 6 is computed from three FMSP runs per size on the same seed-42 pretrained backbone (seeds 42, 43, and 44 at the FMSP/adaptor level); the seed-43 and seed-44

Table 5: Summary of headline results across the registered 1M and 500K models. All values are reproduced from the §6.1–§6.2 master tables; the OOD collapse-inclusive row is from §8.1. Chatter d_2 and the s42 OOD cell are single-seed (seed 42, $n = 1$ backbone); the 3-seed OOD row is the collapse-inclusive mean across FMSP seeds 42/43/44 (the *backbone* is still seed-42-only — only the FMSP adapter was re-run for those three seeds, so the EM row is 3-seed at the FMSP level and $n = 1$ at the backbone level).

metric	V87-1M	V88-1M	V87-500K	V88-500K
chatter d_2 (discord register, s42)	0.883	0.838	0.912	0.738
full-988 EM (3 FMSP seeds; 1 backbone)	563.7	653.3	279.3	396.7
pretrain val PPL	3.18	3.08	3.29	3.20
qrel echo / hijack (s42)	95.0 / 4.0	94.0 / 5.0	37.0 / 49.0	73.0 / 16.0
OOD hijack (registered, s42, collapse-blind)	6.8%	8.5%	35.6%	11.9%
OOD hijack (3-seed mean, s42/43/44; collapse-inclusive)	39.0%	57.6%	39.0%	58.7%
OOD hijack (3-seed mean, collapse-blind)	18.1%	8.5%	35.0%	16.9%

FMSP checkpoints are not separate HF repos but their generation outputs are reproducible from the registered `experiment_10k.py` invocations with `-seed 43` / `-seed 44`. The registered discord-arm repos are `11aa33219/MicroMixer-4-{1M,500K,300K,100K,50K,10K}-Discord-Dialogues` and `11aa33219/MicroT-test1-{1M,500K,300K,100K,50K,10K}-Discord-Dialogues`. The UltraChat, SmolTalk2, and TinyStories [Eldan and Li, 2023] corpus variants are released under the same collections linked in the Code, Data, and Model Availability section.

6 Results

The 3-seed collapse-blind row is the per-seed mean before applying the collapse rule; the 500K cell reads V87 35.0% versus V88 16.9% (3-seed mean; the s42 single-seed reading is 35.6% versus 11.9%, a 23.7pp gap). The two 39.0% means in the V87 collapse-inclusive row (500K and 1M) coincide by averaging, not by sharing a count: the per-seed values differ (500K rescored 39.0 / 37.3 / 40.7 from orig 35.6 / 32.2 / 37.3; 1M rescored 39.0 (per-seed 35.6 / 32.2 / 49.2) from orig 6.8 / 22.0 / 25.4), so two different per-seed distributions happen to average to the same mean — this is a mean-level coincidence, not a duplicated cell. The full per-seed breakdown (V87-500K 35.6/32.2/37.3; V88-500K 11.9/22.0/16.9; V87-1M 6.8/22.0/25.4; V88-1M 8.5/3.4/13.6) is in [OOD_COLLAPSE_RESCORE_ANALYSIS.md](#) §1–§2 and in Appendix B (Table B.1).

The full per-axis master tables follow in §6.1–§6.2; per-size breakdowns at 300K and below are in the same tables.

6.1 The full-988 EM axis: verbatim answer retrieval, not semantic knowledge

Before the tables, the full-988 exact-match axis needs a precise definition to avoid being read as a knowledge test. The fine-tuning corpus (`11aa33219/small-qa-en-10k`, §5.2) provides ten paraphrases of each of 1,000 base questions. For 988 of those base questions, nine paraphrases are placed in the 9,012-example train split ($988 \times 9 = 8,892$) and the tenth is held out; the remaining 12 base questions contribute all ten paraphrases to the train split ($12 \times 10 = 120$), so the train split totals $8,892 + 120 = 9,012$ and the held-out set totals 988 ($9,012 + 988 = 10,000$, matching the corpus size). The 988 held-out questions are therefore the unseen-phrasing paraphrases of 988 of the trained questions, and every held-out question shares its reference answer verbatim with its trained

Table 6: V87 master results across six budgets. WIN bar: $d_2 \geq 0.90$ AND zero cycles. The ‡ symbol denotes degenerate-pass. **Bold = the better of V87 versus V88 at that size on this column** (see Table 6); the two PPL columns (3ep val PPL and Discord PPL) are excluded from the bold policy — V88 wins pretrain val PPL at every size under this resource allocation (§8.8), and the 300K Discord PPL is a 4.47 = 4.47 tie between V87 and V88 (annotated inline). Every column except full-988 EM is a seed-42 single-seed reading ($n = 1$ backbone, since pretraining uses seed 42 only — §5.1); the full-988 EM column is the 3-FMSP-seed mean (seeds 42/43/44) on the seed-42 backbone. (Source: V87 design note §6.1.)

size	params	3ep val PPL	chatter d_2 (cycles)	full-988 EM (3-seed)	qrel echo/hijack	OOD hijack †	fab /18	Discord PPL
1M	996,873	3.18	0.883 (1/9)	563.7 (564/559/568)	95.0 / 4.0 §	6.8% †	18	3.66
500K	491,742	3.29	0.912 (1/9)	279.3 (270/280/288)	37.0 / 49.0 ×	35.6% ×	17	4.00
300K	292,525	3.41	0.810 (1/9)	56.3 (60/60/49)	11.0 / 60.0 ×	25.4% ×	15	4.47 (tie w/ V88)
100K	95,084	3.78	0.546 (9/9)	0.0 (0/0/0)	4.0 / 54.0 ×	27.1% ×	12	4.98
50K	48,684	4.09	0.667 (2/9)	0.0 (0/0/0)	2.0 / 31.0 ×	6.8% ‡	12	5.19
10K	9,666	5.49	0.359 (7/9)	0.0 (0/0/0)	0.0 / 0.0 ‡	0.0% ‡	0 ‡	6.70

twin (full answer overlap). EM on this set measures reproduction of trained answer content under an unseen phrasing of a trained question, that is, verbatim answer retrieval, not semantic knowledge or generalization to new facts. This scoping is consistent with the companion FMSP paper’s finding that memorization under this regime is byte-level pattern matching: the V57 paraphrase-variant EM reads 0 to 1 out of 90 on paraphrases that lie beyond the trained paraphrase set [Lee, Part 2 of this series, companion paper], and with the repetition theory of memorization. Where this paper uses “generalization” as a capability axis (§1), it means paraphrase-form generalization only (same trained content, unseen surface phrasing), never semantic generalization.

6.2 V87 master table (registered)

Auxiliary (seed 42): FT% 71.3, 52.1, 39.8, 33.1, 30.9, 28.4 from 1M to 10K. WO 0.664, 0.402, 0.224, 0.179, 0.166, 0.090. qa-1k TF PPL 1.47, 1.52, 1.60, 2.61, 3.34, 5.68.

† OOD values are the registered single-seed (s42), collapse-blind measurements; see §8.1 for the 3-seed collapse-inclusive rescoring that inverts the 500K ordering. A post-registration 3-seed extension of the discord battery (Part 3, §4.4.1) additionally shows the s42 OOD cells carry large seed noise on this legacy metric: the V87-1M 6.8% rises to an 18.1% 3-seed mean (6.8 / 22.0 / 25.4) and the V87-50K 6.8% to 52.0%, while V88-1M stays at an 8.5% mean — so the s42 1M near-parity does not replicate on the legacy metric, and the 1M OOD comparison must be read collapse-inclusive (§8.1), not from the s42 legacy cells. Binomial SE note for the 1M s42 OOD cells ($n = 59$): V87-1M 4/59 and V88-1M 5/59 each carry SE ≈ 3.3 pp, so the 1.7pp legacy gap (8.5% – 6.8%) is within either cell’s SE — the s42 1M near-parity is also within binomial noise before any seed-noise or collapse-rescore correction.

§ The 1M qrel cell is un-bolded because the echo/hijack split (95.0 / 4.0 versus V88’s 94.0 / 5.0) differs by 1 question out of 100 on each axis, which is within binomial noise at $n = 100$ (SE ≈ 2 percentage points); the cell is reported without a winner mark. The qrel column reports echo/hijack

Table 7: V88 transformer master results across six budgets. Same protocol and seeds as V87. **Bold = the better of V87 versus V88 at that size on this column** (see Table 5); the two PPL columns (3ep val PPL and Discord PPL) are excluded from the bold policy — V88 wins pretrain val PPL at every size under this resource allocation (§8.8), and the 300K Discord PPL is a 4.47 = 4.47 tie with V87 (annotated inline). Every column except full-988 EM is a seed-42 single-seed reading ($n = 1$ backbone, since pretraining uses seed 42 only — §5.1); the full-988 EM column is the 3-FMSP-seed mean (seeds 42/43/44) on the seed-42 backbone. (Source: V88 design note §6.1.)

size	params	3ep val PPL	chatter d_2 (cycles)	full-988 EM (3-seed)	qrel echo/hijack	OOD hijack †	fab /18	Discord PPL
1M	996,736	3.08	0.838 (3/9)	653.3 (663/652/645)	94.0 / 5.0	8.5%	15	3.52
500K	498,528	3.20	0.738 (4/9)	396.7 (386/403/401)	73.0 / 16.0	11.9% †	16	3.98
300K	297,680	3.39	0.840 (2/9)	42.0 (40/43/43)	24.0 / 54.0 ×	18.6%	15	4.47 (tie w/ V87)
100K	97,872	3.77	0.582 (7/9)	0.0 (0/0/0)	8.0 / 74.0 ×	67.8% ×	12	4.92
50K	49,888	4.05	0.553 (5/9)	0.0 (0/0/0)	4.0 / 38.0 ×	23.7%	5 ‡	5.28
10K	9,808	5.48	0.463 (7/9)	0.0 (0/0/0)	0.0 / 0.0 ‡	0.0% ‡	0 ‡	6.79

only; echo + hijack + refuse = 100, and the refuse category (§5.3) is omitted from the cells — it is 1/100 at 1M for both architectures (95.0 + 4.0 = 99; 94.0 + 5.0 = 99) and 0 at every other size shown.

The 1M row is the V83-RPG champion preserved exactly. The 500K row sets the discord-register family chatter record at $d_2 = 0.912$. The 300K row retains anti-collapse chatter ($d_2 = 0.810$, 1/9 cycles) but loses the boundary advantages and most of the memorization.

V76-RPG EM comparison (1M, same chassis class). For honest comparison against the V83/V87 1M row, the V76-RPG backbone (the rotation-pattern parent of V87 and the FMSP chassis on which P05 was developed) under the same P05 recipe reads full-988 EM **544.3 (per-seed 540 / 557 / 536, sd $\approx$ 11.2)** versus V87-1M’s **563.7 (per-seed 564 / 559 / 568, sd $\approx$ 4.5)** — a +19.4 EM mean gain (difference of the displayed one-decimal means; the exact seed-level difference is 19.33). Both 3-seed panels are at the FMSP/adaptor level on a single seed-42 pretrain. The two-sided Welch t-test on the two 3-seed samples gives $t = 2.78$, $df \approx 2.6$, $p \approx 0.08$, which does not clear $\alpha = 0.05$ at $n = 3$, so the EM-axis frontier lift is suggestive rather than significant. A paired t-test on the three per-seed differences (24, 2, 32) gives $t = 2.16$, $df = 2$, $p = 0.16$; the Welch test is the headline because V87’s standard deviation (sd $\approx$ 4.5) is $\sim 2.5\times$ smaller than V76’s (sd $\approx$ 11.2), so the within-group variance is $\sim 6\times$ smaller (variance scales as sd²), and the pairing does not reduce variance here. The deterministic complement is that every V87 seed beats every V76 seed by at least 2 EM (min gap = 559 – 557 = 2). Source: V76-RPG per-seed from paper2 §4.4 and [V77_README.md](#). (Source for V87-1M per-seed: this Table 5.) The V87-versus-V88 discord-1M gap (the registered headline comparison) is now backed by three backbone seeds (§6.8); the V76 comparison in this paragraph remains FMSP-level only.

6.3 V88 master table (registered)

Auxiliary (seed 42): FT% 77.2, 57.1, 44.8, 36.7, 28.8, 28.4. WO 0.764, 0.546, 0.256, 0.198, 0.119, 0.082. qa-1k TF PPL 1.48, 1.54, 1.69, 2.68, 3.30, 5.72.

Table 8: V87 pretrain trajectory per epoch. (Source: V87 design note §3.)

size	params	1ep val PPL	2ep val PPL	3ep val PPL	wall-clock
1M	996,873	3.49	3.39	3.18	~95 min (4070)
500K	491,742	3.59	3.49	3.29	~95 min (4060 Ti)
300K	292,525	3.71	3.61	3.41	~65 min (4060 Ti)
100K	95,084	4.09	4.01	3.78	~45 min (3060 Ti)
50K	48,684	4.41	4.32	4.09	~35 min (3070)
10K	9,666	5.77	5.68	5.49	~12 min (3070)

Table 9: V88 pretrain trajectory per epoch. (Source: V88 design note §3.)

size	params	1ep val PPL	2ep val PPL	3ep val PPL	wall-clock
1M	996,736	3.41	3.23	3.08	~80 min (4070)
500K	498,528	3.51	3.34	3.20	~85 min (4060 Ti)
300K	297,680	3.67	3.50	3.39	~60 min (4060 Ti)
100K	97,872	4.04	3.92	3.77	~42 min (3060 Ti)
50K	49,888	4.37	4.22	4.05	~32 min (3070)
10K	9,808	5.76	5.62	5.48	~11 min (3070)

† OOD values are the registered single-seed (s42), collapse-blind measurements; see §8.1 for the 3-seed collapse-inclusive rescoring that inverts the 500K ordering. A post-registration 3-seed extension of the discord battery (Part 3, §4.4.1) additionally shows the s42 OOD cells carry large seed noise on this legacy metric: the V87-1M 6.8% rises to an 18.1% 3-seed mean (6.8 / 22.0 / 25.4) and the V87-50K 6.8% to 52.0%, while V88-1M stays at an 8.5% mean — so the s42 1M near-parity does not replicate on the legacy metric, and the 1M OOD comparison must be read collapse-inclusive (§8.1), not from the s42 legacy cells. Binomial SE note for the 1M s42 OOD cells ($n = 59$): V87-1M 4/59 and V88-1M 5/59 each carry SE $\approx 3.3\text{pp}$, so the 1.7pp legacy gap (8.5% – 6.8%) is within either cell’s SE — the s42 1M near-parity is also within binomial noise before any seed-noise or collapse-rescore correction. The qrel column reports echo/hijack only; echo + hijack + refuse = 100 (§5.3), and the refuse category is 1/100 at 1M (94.0 + 5.0 = 99) and 0 elsewhere shown.

Under this resource allocation (§8.8), the V88 transformer wins raw language modeling at every size (-0.01 to -0.10 PPL) and raw memorization at 1M and 500K (+16% and +42%). At 300K the V87 mixer wins memorization (56.3 versus 42.0) on the discord register. This 300K mixer win is corpus-dependent and does not replicate across pretraining corpora: Part 3 of this series re-ran the same twelve models on UltraChat and SmolTalk2, and on UltraChat the ordering inverts (V88 211 versus V87 134) while on SmolTalk2 it vanishes (V88 137 versus V87 133) [Lee, Part 3 of this series, companion paper]. The discord-register 300K win is therefore one reading of a corpus-dependent result; the two open explanations (an FMSP-optimizer fork versus a corpus-specific convolution advantage) are discussed in §8.4. At 100K V88 catastrophically fails the OOD boundary (67.8% hijack versus V87’s 27.1%, both collapse-blind; see §8.1). No V88 size passes the WIN bar on chatter. No V87 size strictly clears the registered WIN bar on the discord register either — the 0.912 at 500K and the 0.883 at 1M each carry 1/9 cycles — but the mixer holds a relative advantage (higher d_2 , fewer cycles) without strictly clearing the registered WIN bar; the V87-500K 0.912 family record stands as a number.

6.4 Pretraining dynamics

The V88 minus V87 PPL deltas at 3 epochs are -0.10 , -0.09 , -0.02 , -0.01 , -0.04 , -0.01 from 1M to 10K. Under this resource allocation (§8.8), the transformer wins on raw language modeling at every size, but the margin shrinks with scale-down.

6.5 The scaling patterns

Three scaling patterns hold across both architectures under the P05 recipe. We call these ordering patterns rather than laws: the evidence is two architectures ($n = 2$) and small batteries (9 chatter prompts, 59 OOD prompts, 18 unanswerable questions), so the regularities below are empirical observations on this two-architecture panel, not universal laws of capacity. Part 3 of this series replicates the same ordering across two further pretraining corpora (four architecture-corpus cells: V87 and V88 on UltraChat and SmolTalk2), which is the strongest evidence available [Lee, Part 3 of this series, companion paper].

Pattern 1: memorization cliffs between 500K and 300K and floors at 100K. Full-988 exact-match drops from 563.7 (V87-1M) and 653.3 (V88-1M) to 279.3 and 396.7 at 500K, then to 56.3 and 42.0 at 300K, then to exactly 0.0 at 100K, 50K, and 10K on every seed. Length-stratified exact-match by answer-length quartile (Q1 shortest to Q4 longest, on the registered 988-question battery) follows the same compounding pattern: 1M moves from 69% (Q1) to 47% (Q4), 500K from 37% to 14%, 300K from 10% to 2%, evidence that the failure on longer answers is the dominant component of the compounding cliff above 100K.

Pattern 2 (qrel / near-distribution boundary discipline): boundary discipline on near-distribution unseen questions is a 1M phenomenon for the mixer and a 500K phenomenon for the transformer. V87 qrel hijack 4.0% (1M) to 49% (500K) to 60% (300K); V88 qrel hijack 5% to 16% to 54%. The transformer holds its boundary one budget step further down (at 500K and above, versus at 1M for the mixer). The OOD boundary does not follow the same budget ordering as the qrel boundary: at 100K the transformer fails catastrophically on OOD (67.8% hijack) while the mixer reads 27.1%, inverting the transformer-wins pattern that qrel exhibits at 500K under this allocation (§8.8).

Pattern 3: chatter is the most scale-robust axis on the mixer. V87 chatter d_2 holds at 0.883 (1M), 0.912 (500K), 0.810 (300K) before falling to 0.546 at 100K. The 500K mixer record of 0.912 is the discord-register family best (Part 3’s 0.960 on SmolTalk2 is the project-wide record — §8.4). The chatter axis is not strictly monotone in budget, however: V87 rises from 100K (0.546) to 50K (0.667), and V88 rises from 500K (0.738) to 300K (0.840) — two non-monotonicities that sit alongside the dominant tendency. We state “tendency” rather than “strict order” because of these two inversions.

The ordering pattern that synthesizes the three is: capacity buys the axes in a tendency (not a strict order), fluency then memorization then boundary.

6.6 The per-position CE probe

The per-position cross-entropy probe localizes the attention advantage. The probe was run at the 1M and 500K sizes only; the smaller sizes (300K and below) do not have per-position CE measurements. On the V87-1M and V88-1M backbones, with the 512-item Discord val subset forward-passed once (§5.3), the mean CE stratified by position bucket is:

The attention advantage is band-limited to [64-511]. At long range ([512-1023]) the per-position CE reverses: V88-1M is +0.0047 worse than V87-1M, and V88-500K is +0.0273 worse than V87-500K.

Table 10: Per-position mean CE on Discord val. (Source: V88-vs-V87 analysis §1.)

arch $\times$ size	[0-63]	[64-127]	[128-255]	[256-511]	[512-1023]	overall
V87-1M	1.1115	1.1704	1.1878	1.2238	1.2292	1.2080
V87-500K	1.1424	1.2081	1.2228	1.2533	1.2551	1.2376
V88-1M	1.1078	1.1468	1.1438	1.1714	1.2339	1.1852
V88-500K	1.1438	1.1878	1.1910	1.2133	1.2824	1.2300
Δ (V88 minus V87) 1M	-0.0037	-0.0235	-0.0440	-0.0524	+0.0047	-0.0227
Δ (V88 minus V87) 500K	+0.0014	-0.0204	-0.0318	-0.0401	+0.0273	-0.0076

Table 11: EM-set overlap on 988 held-out questions. (Source: V88-vs-V87 analysis §3.)

size	V87 EM	V88 EM	both	V87-only	V88-only	Jaccard
1M	582	681	498	84	183	0.651
500K	274	398	171	103	227	0.341
300K	60	40	9	51	31	0.099

Table 12: TF compounding on FMSP seed 42. (Source: V88-vs-V87 analysis §2.)

arch $\times$ size	tf_val acc	tf_train acc	acc ¹⁰⁰	all_correct_val	TF-EM-count	greedy EM
V87-1M	0.9934	0.9987	0.518	0.589	582	564
V87-500K	0.9877	0.9930	0.291	0.277	274	270
V87-300K	0.9738	0.9823	0.071	0.061	60	60
V87-100K	0.7715	0.7770	~ 0	0	0	0
V88-1M	0.9956	0.9987	0.645	0.689	681	663
V88-500K	0.9909	0.9944	0.403	0.403	398	386
V88-300K	0.9592	0.9629	0.016	0.040	40	40
V88-100K	0.7478	0.7524	~ 0	0	0	0

By the time context reaches 512+ bytes, $d_{\text{head}} = 16$ with 8 heads (1M) or 6 heads (500K) has exhausted attention’s effective mid-range discrimination.

6.7 EM-set overlap

On the 988-question validation set under teacher-forced decoding:

V88 is not a superset of V87. The 183 V88-only hits at 1M mean the architectures solve different questions in different ways. The Jaccard collapses with capacity because each architecture falls back on its own mechanism when capacity tightens. The qualitative asymmetry suggests V88-only answers reflect content-addressed routing while V87-only answers reflect translation-equivariant pattern completion. The EM gap is a mechanism gap, not a capacity gap, above 300K. Below 300K (100K, 50K, and 10K) the Jaccard is undefined because both architectures score EM 0 on every held-out question, leaving no overlap set to measure.

6.8 TF compounding

Two findings stand out. First, V88 breaks the per-token-independence prediction upward. The per-token-independence prediction is that greedy per-answer EM rate should equal per-token teacher-forced accuracy raised to the answer length if errors were independent (this quantity is denoted

Backbone multi-seed results for the discord-1M cell: the registered seed-42 backbone versus the new seed-43/44 backbones, both architectures, P05 FMSP at seed 42 on every cell. Backbone val PPL and chatter are backbone-level readings; all other rows are post-FMSP. (Source: `logs/bbseed/`.)

arch	metric	s42 (registered)	s43 (new)	s44 (new)
V87	backbone val PPL	3.180	3.179	3.184
	full-988 EM	564	573	561
	chatter d_2 (cycles)	0.883 (1/9)	0.829 (1/9)	0.760 (3/9)
	qrel echo / hijack %	95.0 / 4.0	95.0 / 5.0	90.0 / 10.0
	OOD hijack	4/59 (6.8%)	6/59 (10.2%)	14/59 (23.7%)
	fabrication /18	16.7	17	18
	Discord PPL / qa-1k TF PPL	3.66 / 1.48	3.67 / 1.47	3.68 / 1.46
V88	backbone val PPL	3.080	3.068	3.046
	full-988 EM	663	624	641
	chatter d_2 (cycles)	0.838 (3/9)	0.739 (4/9)	0.701 (3/9)
	qrel echo / hijack %	94.0 / 5.0	93.0 / 5.0	95.0 / 2.0
	OOD hijack	5/59 (8.5%)	3/59 (5.1%)	6/59 (10.2%)
	fabrication /18	16.7	18	17
	Discord PPL / qa-1k TF PPL	3.52 / 1.52	3.49 / 1.50	3.45 / 1.49

acc^{100} in the table below for compactness). At 1M the per-token edge is +0.0022 (0.9956 versus 0.9934), but the TF-EM edge is +99 (681 versus 582) and the all-correct edge is +10 percentage points (0.689 versus 0.589). All-correct is greater than acc^{100} at every size where either model functions, which means within-answer correctness is positively correlated: an answer either snaps into place or misses entirely. That is the empirical signature of coherent retrieval.

Second, 300K inverts. At 300K V88 has lower per-token TF validation accuracy than V87 (0.9592 versus 0.9738), and the lower per-token probability produces a lower EM (40 versus 60). V88-300K train CE is 0.183 versus V87-300K’s 0.104; the transformer has not converged to the FMSP-distribution optimum at this budget.

6.9 Backbone multi-seed: the discord-1M cell at three pretrain seeds

Every backbone in the registered panel above is a seed-42 pretrain, so the backbone-level metrics are $n = 1$ at the pretrain level (only the FMSP adapter is 3-seed — §5.1, Appendix A). In response to the reviewer criticism that the series carries no backbone-level seed replication, we re-ran one cell — discord-1M, the registered headline comparison — at three pretrain seeds. New V87 and V88 backbones were pretrained from scratch at seeds 43 and 44 (checkpoint arms `V87-1M-bb43/-bb44` and `V88-1M-bb43/-bb44`; logs at `logs/bbseed/`), varying the two things a pretrain seed controls: weight initialization and data order. Everything else is held fixed: the P05 FMSP recipe is applied verbatim at FMSP seed 42 on all three backbones of each architecture, so FMSP-level seed noise is held constant and the only varying factor within each architecture is the backbone seed. Each of the six cells then received the full evaluation battery (§5.3). All six runs are new in this revision; the registered seed-42 numbers elsewhere in the paper are unchanged.

Reading. Three findings. (i) The two capability gaps that define the registered 1M comparison replicate across backbone seeds. The transformer is ahead of the mixer on full-988 EM on every backbone seed (per-seed gaps V88 – V87: +99, +51, +80; mean +76.7; means 642.7, sd 19.6, versus 566.0, sd 6.2; two-sided Welch $t = 6.47$, $df = 2.40$, $p = 0.014$ — the $n = 3$ -versus-3 permutation floor is $p \geq 0.10$, so the Welch test is the only test that resolves this gap at $n = 3$) and on pretrain val PPL (V88 ahead by 0.116 on the means: 3.065, sd 0.017, versus 3.181, sd 0.003; Welch $t = 11.55$,

$df = 2.09$, $p = 0.0063$). The registered discord-1M transformer EM advantage is therefore not a backbone-seed artifact. (ii) Chatter is the high-variance axis. The mixer is ahead on all three backbone seeds (per-seed d_2 gaps V87 – V88: +0.045, +0.090, +0.059; mean +0.065), but the mean gap is unresolved at $n = 3$ (Welch $t = 1.19$, $df = 3.93$, $p = 0.30$), and the mixer’s own d_2 drifts down on both new seeds (means 0.824, sd 0.062, versus 0.759, sd 0.071): the registered 0.883 was V87’s favourable draw. No discord-register cell in the table clears the WIN bar ($d_2 \geq 0.90$ AND zero cycles): the registered s42 cells carry 1/9 (V87) and 3/9 (V88) cycles, and the new backbones carry more — on this register the WIN bar has never been strictly cleared (§6.3). (iii) OOD direction confirms the FMSP-level 3-seed reading: V87’s s42 legacy edge (6.8% versus 8.5%) evaporates across backbone seeds (means 13.6% versus 7.9%), the same direction as the registered FMSP-level 3-seed legacy reading (18.1% versus 8.5% — §8.1) and as the collapse-rescore. The qrel bar (echo $\geq 50\%$ AND hijack $\leq 10\%$) is passed by both architectures at 1M on every backbone seed (V87-1M-bb44’s 10.0% hijack passes at the boundary), fabrication is unchanged (17–18/18), and language preservation is stable to ± 0.03 PPL. Scope: this is one cell (discord-1M) at three backbone seeds; every other cell in the panel remains backbone $n = 1$.

7 Ablations and Analysis

7.1 The perfect-ablation property (force_dilation = 0)

The single most informative ablation is the perfect-ablation property described in §4.6. Setting `force_dilation = 0` forces the four-way gate to be one-hot on the `dil = 1` branch, which makes V83/V87 forward pass bit-identical to its RoPE-pattern parent with matched weights. The self-test asserts the maximum difference equal to 0.0 in ablation mode and greater than 10^{-3} with the gate free.

This means three things at once. First, every number the project has ever reported for V76-RPG can be reproduced as the `force_dilation = 0` ablation of V87, which makes the V76 to V87 delta attributable to the content-gated dilation mechanism alone. Second, the V87-1M forward at initialization is bit-identical to a token-mix-zeroed self (the W_o zero-init property), and the four dilated branches contribute zero to the step-0 logits. Third, V76-RPG checkpoints load strict-False into V87-1M with the 21 CCD gate keys missing, which means the architecture is a strict superset of V76 at the parameter level.

The §4.6 fd0 trained-control generalizes this property from the 1M bit-identity corner to a training-from-scratch test at smaller budgets: with the gate removed by *training* rather than by weight-matching, the LTI chassis collapses on chatter (5/9 and 9/9 cycles at 500K and 300K), loses memorization (−45.0 and −24.3 EM), and loses post-FMSP language preservation (roughly $4\text{--}5\times$ Discord PPL) at matched pretrain PPL — so the gate’s measured contribution widens as the budget shrinks, the direction the §3.2 orbit argument predicts. A confound caveat: the fd0 control removes both content-dependence AND long-dilation access — fixing the gate one-hot on `dil=1` shrinks the receptive field from 1024 to 128 positions (§4.1) — so the fd0 collapse is causally undetermined on its own. The uniform-1/4 control of §7.1a resolves this confound by holding content-dependence removed while restoring the four-dilation receptive field, and the verdict (receptive field is the primary anti-collapse ingredient; the content gate is a measured refinement) is reported there.

7.1.1 The uniform-1/4 control: receptive field vs content selection

The fd0 control of §4.6 conflates two changes: removing content-dependent selection (the gate becomes an input-independent constant) *and* shrinking the receptive field to the `dil=1` branch alone.

Table 13: The uniform-1/4 three-way control: CCD vs uniform-1/4 vs fd0 at 500K and 300K (discord register, P05 FMSP, seed 42). uniform-1/4 EM and Discord PPL are single-seed (42); the fd0 EM row is the 3-seed mean with the seed-42 value available for the apples-to-apples contrast. (Source: [UNIFORM_MIXTURE_CONTROL_ANALYSIS.md](#) §5; raw logs in `logs/fu/`.)

budget	chassis	gate	LTI?	RF	chatter d_2 (cycles)	full-988 EM (seed-42 apples-to-apples)	post-FMSP Discord PPL
500K	CCD (registered)	free content gate	no	full	0.912 (1/9)	270 (3-seed mean 279.3)	4.00
500K	uniform-1/4 (this control)	frozen 1/4	yes	full	0.860 (1/9)	273	4.28
500K	fd0 (LTI)	frozen one-hot dil=1	yes	shrunk (128)	0.760 (5/9)	229 (3-seed mean 234.3)	19.96
300K	CCD (registered)	free content gate	no	full	0.810 (1/9)	60 (3-seed mean 56.3)	4.47
300K	uniform-1/4 (this control)	frozen 1/4	yes	full	0.761 (1/9)	24	4.88
300K	fd0 (LTI)	frozen one-hot dil=1	yes	shrunk (128)	0.579 (9/9)	38 (3-seed mean 32.0)	18.23

The uniform-1/4 control separates them. It freezes the CCD gate at the constant uniform mixture $1/K$ at every position (here $K = 4$), so the gate is LTI and input-independent (content-selection removed) while all four dilated branches still fire (receptive field preserved at the full $(k-1) \cdot 8 \cdot N$ reach). The control is implemented as a minimal `force_uniform` hook (`src/model_v83_ccd.py`) that bypasses the gate parameters so they receive zero gradient and stay at zero-init throughout pretrain and FMSP; a forward-check (`verify_uniform_control.py`) asserts gate uniformity 0, all four branches active, input-independence 0, and bit-identity to a free gate with zero parameters. The V76-RPG LTI baseline at these budgets *is* the fd0 row: `force_dilation=0` makes V87 forward bit-identical to its V76-RPG parent (diff 0.0), so a from-scratch fd0 train is a from-scratch V76-RPG train.

The three-way separation isolates the two candidate mechanisms — (H1) content-dependent selection, (H2) the multi-dilation receptive field — and the verdict is axis-dependent:

1. **Anti-collapse (cycle count) → H2 (receptive field).** uniform-1/4 matches CCD *exactly* on cycle prevention (1/9 cycles at both budgets) while fd0 (RF shrunk to dil=1) collapses to 5/9 and 9/9. A wide-RF *LTI* mixture — the uniform gate is input-independent by construction — is sufficient to prevent the periodic-orbit collapse. Content-dependent selection is *not required* for anti-collapse; the active ingredient is the multi-dilation receptive field. This directly tests the §3.2 finite-state framing: an autonomous but wide-RF filter breaks the collapse that an autonomous narrow-RF filter falls into.
2. **Language preservation (post-FMSP Discord PPL) → H2.** uniform-1/4 lands within 0.28–0.41 PPL of CCD (4.28/4.88 versus 4.00/4.47); fd0 degrades 4–5 $\times$ (19.96/18.23). The catastrophic post-FMSP language loss §4.6 reports for fd0 is therefore primarily a receptive-field-shrinkage artifact, not a content-selection-removal artifact: restoring the RF (uniform) recovers language preservation without restoring content-selection.
3. **Chatter fluency (d_2) → mostly H2, plus a smaller H1 bonus.** uniform-1/4 sits between CCD and fd0, nearer CCD (Δ to CCD ≈ 0.05 ; Δ to fd0 ≈ 0.10 –0.18). The content gate buys a modest fluency refinement ($\approx +0.05$ d_2) on top of what the wide RF already delivers.
4. **Memorization (full-988 EM) → budget-dependent (H2 at 500K, H1 at 300K).** This is the one axis that does *not* uniformly favour one hypothesis. At 500K, uniform-1/4 (273) $\approx$ CCD-s42 (270) > fd0-s42 (229) — EM tracks RF. At 300K the ordering inverts to uniform

(24) < fd0 (38) < CCD (60): content-selection is what lifts CCD above both LTI corners, and at this smaller budget fd0’s focused dil=1 filter memorizes more precisely than uniform’s always-mix-4-dilations filter. The gate’s content-dependent routing is EM-positive at small budgets precisely because it can *localize* (pick dil=1 for precision-critical positions) where a fixed uniform mixture cannot. This is the same localization the §7.5 gate-weight table measures directly (dil=1 mass 0.575–0.93 in every block).

Net verdict. The CCD-Mixer’s headline mechanistic story is sharpened, not weakened, by this control. The multi-dilation receptive field is what prevents the periodic-orbit collapse (an LTI uniform mixture at full RF matches CCD on cycle prevention and language preservation); the content gate is a real, measured refinement (Table 13) that lifts chatter d_2 by ≈ 0.05 and restores small-budget memorization (300K EM 60 vs uniform 24) by localizing precision-critical positions. The architecture name CCD-Mixer is unchanged; the content gate is still part of the block and still measured — it is just not the collapse-breaker. The §4.6 reading-point-1 attribution of the fd0 collapse to content-selection removal is, by this cleaner control, primarily an RF-shrinkage artifact.

Caveats. (i) Single seed: the uniform-1/4 arms are FMSP seed 42 only; the fd0 EM reference is 3-seed (42/43/44), and the cleanest apples-to-apples EM contrast is uniform-s42 vs fd0-s42 (reported alongside the fd0 3-seed mean). Chatter, Discord PPL, and pretrain PPL are single-seed for both fd0 and uniform, so those axes are already matched. (ii) Two budgets: only 500K and 300K are run (the two budgets where the fd0 collapse is visible and where §6 shows the LTI-attractor pressure is strongest); extending to 1M and to $\leq 100K$ is the natural follow-up. (iii) Discord register only: like the registered fd0 arms (§8.10), the uniform-1/4 arms are discord-pretrained and discord-evaluated; whether the finding replicates on the UltraChat/SmolTalk2/TinyStories corpora is not measured here.

7.2 NoRemix: the sidecar is robustness, not capacity

The V85 NoRemix ablation deleted the ReMixer sidecar from the V83-RPG chassis, reducing the parameter count from 996,873 to 961,775 (a reduction of 35,098). The result is paradoxical. The NoRemix model produces identical pretrain validation perplexity (3.18, loss 1.1566 versus V83’s 1.1576), better chatter ($d_2 = 0.898$ versus 0.883), and better exact-match (568.3 versus 563.7). On three of four axes the sidecar looks like dead weight. Under the chatter $\leftrightarrow$ EM frontier reading of §3.4–§3.5, V85-NoRemix is in fact *more* above the frontier than V83-RPG (0.898/568.3 versus 0.883/563.7; both are V76-chassis derivatives carrying the CCD gate), but V85 fails the OOD axis (32.2% legacy, 61.0% collapse-inclusive), which is why V83-RPG — the first arm holding all four capability axes above the frontier — remains the champion. On the fourth axis, however, the sidecar is the only thing holding: the NoRemix model fails OOD hijack at 32.2% on the registered legacy (collapse-blind) detector, the same failure signature as the V84-RPG runaway-embedding-stream at 32.2% (19/59).¹ Under the collapse-inclusive rescore of §8.1 (any word 3-gram repeating $\geq 5\times$ reclassified as hijack), V85-NoRemix rises from 32.2% (19/59) to **61.0% (36/59)**, versus V83-RPG’s 6.8% $\rightarrow$ 35.6% (21/59) at seed 42 (rescore run on `logs/v85/eval_ood_v85_samples.txt`). The sidecar’s OOD-insurance value therefore **survives and is amplified** by the corrected metric: the collapse-blind 25.4pp gap (32.2% – 6.8%) becomes a 25.4pp gap at the same seed on the collapse-inclusive reading (61.0% – 35.6%), so the absolute levels shift up but the insurance margin is preserved. The 3-seed V85

¹The two 19/59 counts are produced by two genuinely different mechanisms on two genuinely different architectures (V85 = V83-RPG with the sidecar deleted; V84 = IR-Mixer with a runaway input-stream), not a copy error. Whether the two 19-prompt subsets are literally the same prompts is not registered in the discord-arm eval logs; both architectures saturate the same easy-to-hijack OOD subset, so the shared count is a coincidence of two different failure modes bottoming out on the same subset, not a duplicated measurement.

Table 14: Side-by-side EM readings on the V76/V83 chassis (1M, P05 recipe). (Sources: V80/V82/V83 design notes and Table 5.) The point of this table is that the conditional NcNg contrast (564) sits **above** the registered V87 headline (563.7) but is **not** a chassis-level reading — it is a same-seed, content-clean conditional, and is non-additive with the rotation cost (see §7.3 prose above).

arm	EM	measurement condition
Nc (gate rotation, content clean)	512	V82 conditional contrast on a fixed content-clean background, same seed (not a chassis-level number)
NcNg (both clean)	564	V82 conditional contrast on a fixed content-clean background, same seed (not a chassis-level number; > 563.7 V87 headline)
V82-NcNg (minimal rotation, GCR skip), registered 3-seed	547.3	GCR skip arm, 3 FMSP seeds on the seed-42 V76 backbone
V76-RPG (full rotation), registered 3-seed	544.3	the V76 chassis baseline (P05), 3 FMSP seeds on the seed-42 V76 backbone
V87-1M (CCD-Mixer), registered 3-seed	563.7	the V87 chassis headline (P05), 3 FMSP seeds on the seed-42 V87 backbone

rescore (orchestrator-run, 2026-08-01, on the existing `exp10k-V85-RPG-s43/-s44` FMSP checkpoints) confirms the s42 reading holds across seeds: V85-NoRemix legacy per-seed 32.2 / 23.7 / 30.5 (mean 28.8) rises to collapse-inclusive per-seed 61.0 / 47.5 / 52.5 (mean 53.7), versus V83-RPG’s 39.0 3-seed collapse-inclusive mean (Appendix B, Table B.1) — so the V83-over-V85 OOD championship **holds at 3 seeds** (collapse-inclusive gap 14.7pp, 53.7 – 39.0). The honest championship statement is therefore that V83-RPG is the champion *solely on the OOD axis*: V85-NoRemix dominates it on chatter (d_2 0.898 vs 0.883) and EM (568.3 vs 563.7) at 35,098 fewer parameters, and the only axis on which V83 beats V85 is the OOD axis (where V85 fails both legacy and collapse-inclusive). The sidecar’s function on the champion chassis is pure OOD insurance, and that conclusion is robust to the metric correction and to 3-seed rescoring. The cost is roughly 1.65x training time ($\sim$ 180K versus $\sim$ 109K tokens per second on the RTX 4070).

7.3 The RoPE-on-gate memorization tax

The V82 GCR experiment measured the memorization cost of gate rotation in isolation. With content rotation removed (the Nc endpoint) the same-seed controlled comparison is: Nc (gate rotation, content clean) exact-match 512 versus NcNg (both clean) exact-match 564, a +52 conditional contrast on the content-clean background. The V80 TsR versus TbR arm shows the symmetric finding: a +104 content-rotation contrast. These are conditional contrasts on a fixed background, not a chassis-level tax: registered on the V76 chassis across 3 FMSP seeds, the coherent endpoints are numerically tied (full rotation RPG EM 544.3 versus both-clean NcNg EM 547.3, a +3.0 difference; SD and a paired test were not registered), and the two conditional costs are non-additive — the half-clean Nc endpoint (512) is worse than either coherent endpoint (544.3 and 547.3), so the conditional contrasts cannot be composed into a chassis-level “tax.” The conditional EM effect is a structural consequence of how rotation scatters byte-identity: two byte-identical tokens at different positions produce different representations, which is a noise term in the FMSP write process.

The 564 > 563.7 juxtaposition is therefore explicit but the two numbers are not directly comparable: 564 is a conditional contrast on a fixed background (a same-seed byte-identity-preserved

Table 15: Learned gate weights per block, V83-RPG (mean over positions). (Source: V83 design note §5.3; raw values in `logs/v83/gate_analysis.txt`.) Blocks 4–5 sit inside the “block 1 to 3” band shown here (dil=1 mass 0.86 to 0.93) and are collapsed into that row to avoid clutter; the full per-block table from 9,666 parameters to 996,873 parameters is in [V87_SCALING_ANALYSIS.md](#). The point estimate for every cell is the per-block mean over positions; the ranges span the blocks within each group (the single-block rows report the within-block min-max across the gate-weight file’s per-position bins before the per-block mean is taken; the multi-block “1 to 3” row reports the across-block min-max of the per-block means).

block	dil=1	dil=2	dil=4	dil=8	τ
0	0.575 to 0.609	0.196 to 0.221	~ 0.09	~ 0.11	1.5
1 to 3	0.86 to 0.93	~ 0.05	~ 0.02	~ 0.02	1.1 to 2.0
6	0.618 to 0.627	0.217 to 0.220	~ 0.09	~ 0.07	3.7 to 3.9

write), while 563.7 is a chassis-level 3-seed mean under the standard P05 recipe. The non-additivity (the half-clean Nc endpoint at 512 is worse than either coherent endpoint) is the formal reason the conditional NcNg lift cannot be turned into a chassis-level tax or a chassis-level lift.

7.4 CCD does not replace RoPE

The V83-NcNg arm is the empirical demonstration. With gate rotation removed on the V83 chassis, chatter distinct-2 regresses to 0.733, below the V82-NcNg baseline of 0.791. CCD without full rotation makes chatter worse, not better. The finite-state framing of §3.2 explains why: the content gate alone is still an autonomous map on a finite state space, so it is still eventually periodic, and the §7.1a uniform-1/4 control confirms an LTI but wide-RF filter matches CCD on cycle prevention — the gate is not the collapse-breaker. The joint endpoint works because two independent ingredients compose: RoPE’s absolute-position rotation is the non-autonomous term that removes the periodicity guarantee, and the multi-dilation receptive field provides the anti-collapse width that a narrow-RF LTI filter (fd0) lacks; the content gate is the measured refinement that lifts chatter d_2 by ≈ 0.05 and small-budget EM by localizing precision-critical positions (§7.1a). The conservation pattern is Pareto-dominable only at this joint endpoint.

7.5 The learned dilation schedule

The V83 trained gate weights, averaged over positions, show a U-shaped schedule that emerged from data rather than from a hard-coded hierarchy:

The schedule is U-shaped, not the classic “local low-level, global high-level” hierarchy: the dil=1 branch dominates in *every* block (mass 0.575 to 0.93), and dil=8 carries ≤ 0.11 mass anywhere, so no block shifts its mass toward the long dilations that a local-to-global hierarchy would require. Block 0 starts soft (dil=1 mass 0.575 to 0.609, temperature 1.5); blocks 1–5 sharpen to near-one-hot on dil=1 (mass 0.86 to 0.93); block 6 softens again (dil=1 mass 0.618 to 0.627, temperature rising to 3.7 to 3.9). The mass-weighted effective receptive field per block is therefore far below the 1024-position maximum: block 0 ≈ 273 positions, blocks 1–3 ≈ 159 positions, block 6 ≈ 253 positions (e.g. block 0 $\approx 0.60 \cdot 129 + 0.21 \cdot 257 + 0.09 \cdot 513 + 0.10 \cdot 1025$), so the gate spends almost all of its mass on the local view and only a small fraction on long-range context. This connects directly to the §7.1a control: the gate’s measured preference for dil=1 is exactly the localization that wins 300K EM (CCD 60 vs uniform 24), because the content gate can pick dil=1 for precision-critical positions where a fixed uniform mixture always mixes in the noisier long dilations. The same U-shape is observed at every

Table 16: V86 census at 1M parameters, P05 recipe, 3 seeds. (Source: V86 design note §1.)

family	version	params	EM (3-seed)	chatter d_2	qrel/OOD jack	hi- jack
RPG-Mixer (champion)	V83-RPG	996,873	563.7 (564/559/568)	0.883	4.0% / 6.8%	
RPG-Mixer minus ReMixer	V85-NoRemix-RPG	961,775	568.3 (575/554/576)	0.898	5.0% / 32.2% ×	
MLP-Mixer + ReMixer (learned labels)	V52-P05	976,398	560.7 (575/554/553)	0.866 (1/9)	18.0% × / 15.3% ✓	
MLP-Mixer + ReMixer (paper-faithful)	V58-P05	962,055	455.0 (466/448/451)	0.748 (4/9) ×	22.0% × / 5.1% ✓	
+ saliency	V59-P05	962,958	436.0 (430/441/437)	n/m	n/m	
Pure MLP-Mixer	V46-3-P05	961,152	431.3 (414/438/442)	n/m	n/m	
FSC-Mixer + sidecar	V80-TsR	995,248	488 (s42)	0.732	30.0% × / 60 to 80% ×	
FSC-Mixer, pure	V86-TsR	955,136	376.7 (371/363/396)	0.774	34.0% × / 49.2% ×	

preset (the V87 scaling analysis documents the recapitulation from 9,666 parameters to 996,873 parameters exactly), which suggests the schedule is data-driven.

7.6 The V86 census

V83-RPG is the only arm holding all four axes. The old “routing ceiling” was a recipe artifact: every old-era byte-AR family gained hundreds of exact-match under P05 (V46-3 rose to 431.3, V58 rose to 455.0, V59 rose to 436.0), and the ceiling became visible only because the new recipe pushed the old arms into the regime where their ceilings could be measured.

8 Limitations

This section catalogs the post-registration caveats that the paper must carry. None change the registered architecture, recipe, or headline numbers, but each either reframes a registered result or bounds its interpretation.

8.1 OOD hijack legacy numbers are collapse-blind and single-seed-noisy

The registered OOD hijack rates were measured on a single seed (42) against a discord-register-matched battery (here and throughout, “register” denotes the style of the training distribution, e.g. casual discord conversation versus formal instruct). A periodic-orbit collapse matches no memorized answer, so the hijack detector scores it “free.” The collapse-rescore analysis (probe A: any word 3-gram repeating at least five times in the generated output equals collapsed equals reclassify as hijack) reverses the 500K ordering. Under 3-seed collapse-inclusive rescoring, the 500K boundary reads V87 39.0% versus V88 58.7%, and the 1M near-parity reads V87 39.0% versus V88 57.6%. The collapse correction exceeds the measured gap by a factor of about 2.1: a 37.8 percentage-point swing on an 18.1 percentage-point measured gap (the 3-seed mean gap, V87 35.0% minus V88 16.9%, from [OOD_COLLAPSE_RESCORE_ANALYSIS.md](#) §1; the seed-42 single-seed gap is 23.7pp, V87 35.6% minus V88 11.9%). This 18.1pp *gap* is unrelated to the 18.1% 3-seed OOD *rate* cited in the Table 5/6 † footnotes (the V87-1M 6.8% rising to an 18.1% 3-seed mean) — the two “18.1” numbers are a

coincidence: one is a difference of two 500K rates, the other is a 1M rate. So the 500K V88 OOD “advantage” in the registered table was a collapse artifact rather than a genuine boundary win.

The legacy metric is also seed-noisy on plain re-measurement, before any collapse rescoring. A post-registration 3-seed extension of the discord battery (Part 3, §4.4.1) re-ran the identical legacy detector on seeds 43/44: V87-1M rises from the s42 6.8% to an 18.1% 3-seed mean (6.8 / 22.0 / 25.4), V87-50K from 6.8% to 52.0%, while V88-1M stays at an 8.5% mean (8.5 / 3.4 / 13.6) — the s42 1M near-parity does not replicate on the legacy metric, and at 3 seeds the transformer leads on that metric at 1M. The mixer’s 1M boundary claim therefore stands on the qrel and collapse-inclusive axes, not on the legacy s42 OOD cell.

8.2 OOD is a register-pair property

The mechanism analysis decomposed the OOD hijack rate into a probe-anchor component (roughly 2 to 3x) and an eval-register component (roughly 2 to 3x). On a formal-instruct OOD battery, the discord-pretrained arms themselves read 41.7% (V87) and 18.3% (V88) hijack. OOD for this project must be reported as a register-pair property, not a model property.

8.3 The chatter advantage is register-dependent

The dataset-comparison analysis [Lee, Part 3 of this series, companion paper] (companion paper, Part 3 of this series) re-ran the same twelve models on SmolTalk2 and observed that the transformer clears the WIN bar at both 1M (0.911) and 500K (0.932); the mixer also clears the WIN bar on SmolTalk2 (0.960 at 1M and 0.900 at 300K), while setting the new family record on that corpus. On the discord register, by contrast, no V87 size strictly clears the registered WIN bar (the 0.912 at 500K and the 0.883 at 1M each carry 1/9 cycles), so the discord chatter result is a relative advantage (higher d_2 , fewer cycles) rather than a strict WIN-bar pass. The V87-500K 0.912 on the discord register is a real and reproducible number, but it is not a universal property of the architecture.

8.4 The V87-300K EM win over V88 has an unresolved capacity-vs-optimizer fork, and is corpus-dependent

The V87-300K exact-match is 56.3 versus V88-300K 42.0 on the discord register, with V88-300K train CE 0.183 versus V87-300K’s 0.104. Two readings are possible: convolutional weight sharing genuinely buys more write-capacity per parameter at small d , or the fixed FMSP learning rate 1×10^{-3} is too hot for the 5-head attention at 300K. The two readings are not distinguished by the registered discord data, and the optimizer-fork reading remains one of two open explanations. The third, decisive piece of evidence comes from Part 3 of this series: the same twelve models re-pretrained on UltraChat and SmolTalk2 show that the discord-register 300K mixer win does not replicate across corpora. On UltraChat the ordering inverts (V88-300K 211 versus V87-300K 134) and on SmolTalk2 it vanishes (V88-300K 137 versus V87-300K 133) [Lee, Part 3 of this series, companion paper]. The discord-register 300K mixer win is therefore corpus-dependent, not a universal property of convolution-versus-attention at small d ; whether it reflects an FMSP-optimizer interaction specific to the discord register or a genuine but corpus-specific convolution advantage is unresolved by the available data.

Self-containment caveat. Two corrections carried in this paper originate in Part 3 and are reproduced here for reviewer convenience, but Part 3 should be loaded to verify them at source: the 3-seed OOD extension cited in the Tables 5/6 † footnotes and in §8.1 (V87-1M 6.8% $\rightarrow$ 18.1% 3-seed mean; V87-50K 6.8% $\rightarrow$ 52.0%; V88-1M 8.5% mean) is from Part 3 [Lee, Part 3 of this series, companion paper, §4.4.1]; and the SmolTalk2 register comparison that re-frames the chatter moat as

register-dependent (§8.3, §8.4 above) is from Part 3 [Lee, Part 3 of this series, companion paper, §4.7]. The discord-register headline numbers in Tables 5 and 6 are Part 1’s own and are reviewer-verifiable without Part 3; only the two cross-paper corrections above carry that dependency.

8.5 The 500K chatter advantage is suggestive, not registered

V87-500K chatter is 0.912, above V87-1M’s 0.883. The chatter battery has nine prompts, and no prompt-set confidence interval was registered for the 9-prompt battery; treat the 0.029 distinct-2 gap as an estimate, not a measured band.

8.6 Safety asymmetry: V88 fails loudly, V87 fails fluently

The qrel hijack set on the 500K models reveals an asymmetry the registered metric does not capture ($n = 49$ hijack samples, single-annotator classification). Hand-classification: V88-500K hijacks are approximately 8% verbatim-under-wrong-key, approximately 72% corrupted text, and approximately 20% other (mixed/uncategorizable); V87-500K hijacks are approximately 52% verbatim-under-wrong-key, approximately 36% corrupted, and approximately 12% other. V88 fails loudly (corrupted text is easy for a downstream filter or a human reader to detect), while V87 fails fluently (clean verbatim facts under wrong keys pass every detector). The V87 failure mode is epistemically more dangerous.

8.7 What was not measured

Two measurements would strengthen the paper and are not in the registered data: a learning-rate sweep at V88-300K, and an exposure-bias correction on the EM counts. (Length-stratified exact-match by answer-quartile, reported in §6.4, is part of the registered scaling analysis.)

8.8 The V88 reference is one resource allocation, not attention in general

The V88 transformer is deliberately boring (§3.7, §4.7), and its specific allocation choices confound the mixer-versus-transformer comparison. At 1M the V88 reference uses $d_{\text{head}} = 16$, 5 blocks (versus the mixer’s 7), a GELU MLP (versus the mixer’s SwiGLU), and no biases anywhere. The comparison in §6 therefore measures THIS resource allocation (shallow, GELU, bias-free, narrow-head attention) against the mixer’s allocation, not attention-versus-convolution in general. Two downstream readings inherit this confound: the band-limited attention advantage in §6.5 (attention wins on mid-range positions but loses at long range) could reflect the narrow $d_{\text{head}} = 16$ exhausting its discrimination budget rather than a fundamental limit of attention; and the mechanism-gap reading of the EM-set overlap in §6.6 (V88-only answers as content-addressed routing) could partly reflect GELU-versus-SwiGLU channel-mix differences rather than a pure attention mechanism signature. A depth-matched or FFN-matched transformer control (same block count and SwiGLU channel-mix as the mixer, with only the token-mix swapped) is the obvious strengthener and is future work. This paper’s claims are scoped to the registered V87-versus-V88 comparison under these specific allocations.

8.9 All generation metrics are measured under one decode configuration

As disclosed in §5.3, every generation-based metric in this paper (chatter distinct-2, full-988 exact-match, question-relevance, OOD, unanswerable fabrication) is measured under a single registered decode configuration: greedy decoding (temperature 0) with repetition penalty 1.2 and a no-repeat-4-gram constraint. In this codebase, both knobs are applied only inside the sampling branch and are never reached under temperature 0, so the greedy branch takes argmax on unmodified logits

and the registered repetition penalty 1.2 and no-repeat-4-gram are inert. Raw-greedy verification on all 24 pretrained backbones (12 discord + 12 corpus-sweep) confirms byte-identical outputs to the registered runs, so the anti-collapse numbers (chatter scores, cycle counts, and the collapse-rescored OOD readings in §8.1) are properties of unassisted greedy decoding, not of any decoder-assisted variant. The configuration is applied identically to both architectures, so the cross-architecture comparisons remain fair; this is a measurement scoping rather than a registered weakness of either architecture.

8.10 The fd0 trained-control arms are discord-only and single-seed outside EM

The fd0 trained-control study in §4.6 (and the §7.1 pointer) is scoped in two respects. First, the fd0 arms are pretrained and fine-tuned on the discord register only; the dataset-sweep corpora of Part 3 (UltraChat, SmolTalk2, TinyStories) were not re-run with `force_dilation=0`, so whether the post-FMSP language-preservation collapse (Table 2, reading point 3) replicates on the other corpora is not measured here. Second, all non-EM fd0 metrics (chatter, qrel, OOD, language preservation, unanswerable fabrication) are the registered single-seed (s42) measurements, whereas the full-988 EM row is 3-seed (42/43/44); the four-point reading is therefore a discord-register, partially-single-seed reading. Extending the fd0 control to the other corpora and to 3-seed boundary axes is the natural next step.

8.11 Parameter-matching is not compute-matching

The CCD token-mix applies its shared $k = 129$ depthwise kernel at four dilations (1, 2, 4, 8) per position, then binds the four views through the per-position gate. The four dilated applications of the same kernel cost roughly $4\times$ the conv FLOPs of a single-view dilated-convolution token-mix at the same parameter count, so the V87-versus-V88 comparison in §6 is parameter-matched but not conv-FLOP-matched. No throughput or FLOP count is registered in this paper for either architecture; the only registered throughput number is the §7.2 NoRemix training-cost figure ($\sim 180\text{K}$ versus $\sim 109\text{K}$ tokens per second on the RTX 4070), which is a V83-with-sidecar versus V85-without-sidecar contrast, not a V87-versus-V88 contrast. A FLOP-matched or throughput-matched V87-versus-V88 control (for example, halving the mixer’s block count or kernel span to roughly offset the $4\times$ dilation cost) is the natural strengthener and is future work.

8.12 Winner’s curse / post-selection

V87 was selected from approximately 80 candidate architectures (V9–V88) evaluated on variants of the same eval battery; the headline V87-vs-V76 comparison is therefore post-selection, and the p-values reported in §3.5 and §6.1 (Welch $p \approx 0.08$, paired $p \approx 0.16$) carry no multiple-comparison correction across the selection pipeline. The perfect-ablation property (§4.6) partially mitigates this concern by making V87-vs-V76 a controlled mechanism contrast rather than an arbitrary post-hoc comparison: with `force_dilation=0`, the V87-1M forward is bit-identical to the V76-RPG parent, so the V87-vs-V76 EM delta isolates the CCD gate’s effect rather than reflecting a cherry-picked contrast among unrelated arms. The broader selection across V9–V88 nonetheless remains a winner’s-curse risk that the reader cannot fully assess from the single registered comparison alone.

8.13 External capability anchor (WikiText-103 byte PPL)

No external capability benchmark was run for the registered study; every capability number in the registered tables derives from the project-internal batteries (chatter, full-988, qrel, OOD, fabrication,

Table 17: Post-registration external anchor: WikiText-103 (raw) test byte-level perplexity of the pretrained V87/V88 backbones (pre-FMSP). **Bold = the better of V87 versus V88 at that size.** All cells are measured on the full 1,292,013-byte test stream. GPT-2-small (124M parameters) is the reference scorer, BPB only.

size	arch	byte-PPL	BPB
1M	V87	6.66	2.74
1M	V88	7.87	2.98
500K	V87	7.09	2.83
500K	V88	7.32	2.87
300K	V87	7.37	2.88
300K	V88	7.98	3.00
100K	V87	8.32	3.06
100K	V88	9.06	3.18
50K	V87	9.09	3.18
50K	V88	10.11	3.34
10K	V87	13.64	3.77
10K	V88	14.83	3.89
GPT-2-small (124M)		—	1.25

Discord PPL) on the Discord-Dialogues / small-qa-en-10k corpora. Post-registration, we compute the natural cross-paper calibration: byte-level perplexity of the pretrained backbones (pre-FMSP, discord-Dialogues, 3 epochs) on the official WikiText-103 raw test split (4,358 lines, 1,292,013 UTF-8 bytes), scored as contiguous 1023-byte blocks at ctx 1024 under the same mean-CE-per-byte reduction as Part 3’s TinyStories anchor (`eval_wikitext_anchor.py`; full log `logs/eval_wikitext_anchor.log`), with GPT-2-small (124M parameters) as the reference scorer in bits per byte ($\text{BPB} = \text{CE} / \ln 2$; BPE tokens are not bytes, so only BPB is comparable).

Three readings. First, the absolute calibration: both families sit far above GPT-2-small (BPB 1.25 against 2.74–3.89) — the expected gap at two-plus orders of magnitude fewer parameters — and far above their own discord validation PPL (≈ 3.1 – 3.2): the domain shift from Discord-Dialogues pretraining to formal Wikipedia text costs both architectures roughly four perplexity points. Second, the ordering inverts: on this external formal corpus the mixer reads lower than the transformer at all six budgets (1M: 6.66 versus 7.87), so the transformer’s every-size raw-LM win of §6.3 is a property of the pretraining register, not of the architectures — the same register-dependence Part 3 established for the chatter and memorization axes, now observed on text neither model was trained on. Third, the MMLU/HellaSwag/LAMBADA class of capability benchmarks remains unrun: those suites presuppose subword-scale semantic competence that this parameter regime does not approach, so the byte-PPL anchor is the external calibration that is meaningful at this scale.

8.14 The untied 4-kernel arm was not run

The CCD block’s four branches are dilated samplings of one shared depthwise kernel — a strictly *more constrained* parameterization than tying four arbitrary expert kernels, since tying four arbitrary DynamicConv experts to a single weight matrix would lose the dilation/input-stride structure and would not reproduce CCD. An untied 4-kernel variant (four separate per-dilation kernels, content-gated, at roughly $4\times$ the conv parameters) was not run, so the shared-kernel constraint’s contribution to the measured CCD effect is unmeasured. This is a confound on top of the §7.1a receptive-field-versus-content-selection separation: the uniform- $1/4$ control isolates the gate’s content-dependence from the receptive field, but the shared-kernel constraint itself (tying the four dilated kernels into

one tensor) is never broken in any registered arm. Mechanistically, the shared kernel forces the four dilated views to be the same filter sampled at different strides, so the CCD block cannot learn four independent per-dilation filters; whether that constraint is the right inductive bias for the anti-collapse purpose is open. An untied 4-kernel arm (or a partially-untied arm, e.g. tying the small-dilation pair and untied for the long-dilation pair) is the natural strengthener and is future work. The CCD-Versus-V76 comparison in this paper therefore conflates two ingredients: the content gate (isolated by §7.1a) and the kernel-sharing constraint (not isolated); readers should not interpret the +19.4 EM lift, the chatter $d_2 = 0.912$ record, or the OOD reading as evidence specific to the shared-kernel parameterization alone.

9 Conclusion

The CCD-Mixer is a content-gated mixture of shared-weight dilated convolutions, designed for byte-level autoregressive language modeling under one million parameters, with no attention, recurrence, or state-space machinery. The token-mix is non-LTI (its effective filter is a function of the input), but this alone does not prevent the periodic-orbit collapse that pure LTI mixers fall into: on a periodic orbit the gate’s input is periodic, so an autonomous content-gated map preserves P-periodic dynamics. The collapse is removed at the joint endpoint, where RoPE supplies the non-autonomous forcing that breaks the autonomous-map guarantee and the multi-dilation receptive field provides the anti-collapse width; the content gate is a measured refinement (chatter +0.05 d_2 , small-budget EM by localization — §7.1a), not the collapse-breaker, and the V83-NcNg negative control confirms the gate without rotation regresses chatter (0.733). The block adds 517 parameters per block to a 128-dim backbone.

The six-budget scaling study shows that capacity buys the four capability axes in a tendency (fluency then memorization then boundary; the chatter axis is not strictly monotone — §6.4). Fluency holds to about 300K parameters on both architectures (a relative advantage; no discord-register size strictly clears the registered WIN bar of $d_2 \geq 0.90$ with zero cycles — §6.2, §8.3); memorization requires 300K and falls off a cliff between 500K and 300K with a hard floor at 100K; boundary discipline is a 1M phenomenon for the mixer and a 500K phenomenon for the vanilla transformer. The headline results are a 0.912 distinct-2 chatter score at 500K parameters (the discord-register family record; Part 3’s 0.960 on SmolTalk2 is the project-wide record — §8.4) and 563.7 full-988 exact-match memorization at 1M parameters across three seeds.

The honest framing is that, under this resource allocation (§8.8), the transformer wins raw language modeling at every size (-0.01 to -0.10 PPL), wins raw memorization at 500K and above (+16% at 1M, +42% at 500K), and wins the registered single-seed boundary metrics at 500K, while the mixer leads them at 1M (on the registered single-seed cells; the 3-seed legacy OOD inverts this — §8.1). The mixer wins anti-collapse chatter fluency on the discord register, wins mid-size robustness at 300K (a corpus-dependent result; see §8.4), and wins the OOD boundary at 1M under collapse-inclusive rescoring. The scaling patterns are consistent across both architectures we tested ($n = 2$); the chatter advantage is register-dependent; the safety asymmetry reframes the boundary axis as a qualitative choice, not a quantitative one.

This is a study at the sub-million-parameter scale, where every architectural, recipe, and measurement choice can be held fixed across a six-budget panel: that is the source of the controlled comparison, not an apology. The contributions are five portable lessons grounded there: a mechanistic account of attention-free repetition collapse (§3), a measured chatter↔memorization conservation pattern and the multi-dilation primitive that, composed with rotation, lifts above it (the +19.4 EM lift is suggestive rather than significant at $p \approx 0.08$, $n = 3$; §3, §4, §6.1, §7.1a), capability-ordering

patterns consistent across both architectures tested (§6), and a post-registration audit trail that corrects its own headline numbers (§5, §8).

Architecture-comparison findings in this paper describe this two-architecture panel only; comparisons against other non-attention backbones (state-space models such as Mamba [Gu and Dao, 2023], gated long convolutions such as Hyena [Poli et al., 2023]) are future work.

10 Code, Data, and Model Availability

All source code, the V9 through V88 design-note archive, the self-test batteries, and the orchestration scripts are public under Apache-2.0 at <https://github.com/llaa33219/MicroMixer-4>. All trained checkpoints are released as safetensors-only Hugging Face repos under two collections: <https://huggingface.co/collections/llaa33219/micromixer-4> (V87 family) and <https://huggingface.co/collections/llaa33219/microt-test1> (V88 family), with repos named `llaa33219/MicroMixer-4-{1M,500K,300K,100K,50K,10K}-{Discord-Dialogues,UltraChat,SmolTalk2,TinyStories}` and `llaa33219/MicroT-test1-{size}-{corpus}` respectively (the corpus suffix is part of the repo name). The pretraining corpus is `mookiezi/Discord-Dialogues` and the FMSP dataset is `llaa33219/small-qa-en-10k` on Hugging Face.

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Companion papers in this series:

Lee, S. The Objective Is the Lever: FMSP and the Composite-Objective Account of Fine-tuning Sub-1M Language Models. Companion paper, Part 2 of this series.

Lee, S. Corpus First at the 1M Budget: A Controlled Pretraining-Corpus Sweep of Sub-1M Mixer and Transformer Language Models. Companion paper, Part 3 of this series.

Appendix A. Scope reminders

- All architecture-comparison claims in this paper are scoped to the registered V87 (CCD-Mixer) versus V88 (vanilla transformer) pair under their specific resource allocations (§8.8). The backbone pretraining is seed-42-only (§5.1) outside the discord-1M cell, which is now reported at three pretrain seeds (§6.8); for every other cell, every backbone-level metric (chatter distinct-2, qrel echo/hijack, OOD hijack, fab, Discord PPL) remains an $n = 1$ reading. Only the full-988 exact-match column is 3-seed, and that 3-seed mean is at the FMSP/adaptor level (seeds 42/43/44) on the single seed-42 pretrained backbone — outside §6.8 it is not a 3-backbone-seed mean.

- No new experiments were run for the revision that added those items: every number added in the §8.1 disambiguation, the Table 4 3-seed rows, and the Appendix B table already exists in the project’s post-registration analysis notes ([OOD_COLLAPSE_RESCORE_ANALYSIS.md](#), [V87_SCALING_ANALYSIS.md](#), [V88_VS_V87_ANALYSIS.md](#)).

Appendix B. OOD hijack, 3 seeds $\times$ 2 batteries $\times$ 2 sizes (collapse-rescore)

Procedure (probe A, summary). For each OOD generation, count word-level 3-grams in the output; any 3-gram repeating $\geq 5\times$ is flagged as a periodic-orbit collapse and the output is reclassified as hijack (the rule is purely lexical, applied to the existing samples files, with zero training or parameter updates). Threshold-robustness sweeps the cutoff from $\geq 3\times$ through $\geq 10\times$; the 500K inversion is stable from $\geq 3\times$ to $\geq 8\times$ and only inverts at the most extreme cutoff ($\geq 10\times$). The 3-seed, 2-size, 2-battery matrix below is the registered rescore; the full causal narrative and the $\approx 209\%$ artifact quantification live in [OOD_COLLAPSE_RESCORE_ANALYSIS.md](#) §1.

This appendix surfaces the per-seed OOD hijack readings that underlie the 3-seed means cited in Table 4 and §8.1, so the 500K inversion and the 18.1pp / 23.7pp gap readings are reviewer-verifiable inside Part 1 without loading [OOD_COLLAPSE_RESCORE_ANALYSIS.md](#). The 500K row of this table also appears in Part 3 (§5.6), but Part 3 reports only the 500K 3-seed mean in-table (per-seed rescored triplets in prose); the full per-seed, per-size rescore table is unique to this appendix and to [OOD_COLLAPSE_RESCORE_ANALYSIS.md](#). The source-of-truth file is [OOD_COLLAPSE_RESCORE_ANALYSIS.md](#) §1–§2. “orig” = the registered collapse-blind detector; “coll” = collapsed outputs newly flagged as hijack that were *non-hijack at orig* (so orig-hijack + coll = rescored-hijack arithmetically for every row); “rescored” = hijack count after the collapse rule (any word 3-gram repeating $\geq 5\times$). Note on the coll column versus the Parts 2–3 figure: the coll column here counts newly-flagged non-hijack collapses (V88-500K per-seed 27/22/25, mean 24.7), whereas the “25.7 of 59” figure quoted in Parts 2–3 is the mean of TOTAL collapsed outputs including already-hijack ones (27/25/25, mean 25.7); the two definitions differ only for the s43 seed (22 here versus 25 there). Casual battery = `ood_eval_v77.txt` ($n = 59$). Backbones are seed-42-only; the three seeds are FMSP/adaptor seeds 42/43/44 on those seed-42 backbones. The V87-1M s43 cell was re-run on 2026-08-01 after the original `logs/v87/eval_ood_v87-s43_samples.txt` was found to be a 0-byte file; the regenerated file produced 19/59 rescored hijack (32.2%), and the V87-1M 3-seed mean of 39.0 (per-seed 35.6 / 32.2 / 49.2 from orig 6.8 / 22.0 / 25.4) is built from the regenerated s43 cell.

Reading the 500K cell as (V87 hijack% – V88 hijack%): the 3-seed mean gap is +18.1pp collapse-blind (V87 35.0% – V88 16.9%) and –19.7pp collapse-inclusive (V87 39.0% – V88 58.7%), a 37.8pp swing. The seed-42 single-seed gap is +23.7pp collapse-blind (35.6% – 11.9%). The two “18.1” numbers in this paper are distinct: the 18.1pp *gap* above (a 500K difference), and the 18.1% V87-1M 3-seed OOD *rate* (6.8/22.0/25.4 mean) cited in the Tables 5–6 † footnotes — they coincide numerically by chance.

Table 18: OOD hijack per seed, casual battery ($n = 59$).

arm	seed	orig echo/hijack/free	orig hijack%	coll	rescored hijack%
V87-500K	s42	4 / 21 / 34	35.6	2	39.0
V87-500K	s43	5 / 19 / 35	32.2	3	37.3
V87-500K	s44	1 / 22 / 36	37.3	2	40.7
V87-500K	mean		35.0	2.3	39.0
V88-500K	s42	9 / 7 / 43	11.9	27	57.6
V88-500K	s43	9 / 13 / 37	22.0	22	59.3
V88-500K	s44	10 / 10 / 39	16.9	25	59.3
V88-500K	mean		16.9	24.7	58.7
V87-1M	s42	—	6.8	17	35.6
V87-1M	s43	—	22.0	6	32.2
V87-1M	s44	—	25.4	14	49.2
V87-1M	mean		18.1	12.3	39.0
V88-1M	s42	—	8.5	26	52.5
V88-1M	s43	—	3.4	28	50.8
V88-1M	s44	—	13.6	33	69.5
V88-1M	mean		8.5	29.0	57.6